

Electrical Engineering Department

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Courses offered by the Department of Electrical Engineering are listed under the subject code EE on the *Stanford Bulletin's* ExploreCourses website.

The Department of Electrical Engineering (EE) at Stanford innovates by conducting fundamental and applied research to develop physical technologies, hardware and software systems, and information technologies; it educates future academic and industry leaders; and it prepares students for careers in industry, academia, and research labs.

Electrical Engineering has effected societal changes at the heart of the information revolution. Electrical and electronic devices—realized in both hardware and software—are integral to daily life, whether in the home, in health care, in recreation, or in the infrastructure for communication and computation. Electrical engineers use theories and tools from mathematics and physics to develop systems ranging from smart electric grids, wired and wireless communications and networking, embedded systems, integrated electronics, imaging and sensing devices, to Internet-based information technology.

The Electrical Engineering Department offers the following degrees: Bachelor of Science, Master of Science, and Doctor of Philosophy. The department also offers joint degrees in Electrical Engineering and Law (M.S./J.D.) and Electrical Engineering and Business Administration (M.S./M.B.A.). A minor can be obtained for the Bachelor of Science and Doctor of Philosophy.

Disciplinary Areas in Electrical Engineering

Research in Electrical Engineering spans a diverse set of intellectual disciplines and applications. The disciplines can be grouped into three overlapping and interrelated areas:

Hardware/Software Systems

- Data Science
- Embedded Systems
- Energy-Efficient Hardware Systems
- Integrated Circuits and Power Electronics
- Mobile Networking
- Secure Distributed Systems
- Software Defined Networking

Information Systems and Science

- Biomedical Imaging
- Control and Optimization
- Data Science
- Information Theory and Applications
- Machine Learning
- Societal Networks
- Signal Processing and Multimedia

Physical Technology and Science

- Biomedical Devices, Sensors and Systems
- Energy Harvesting and Conversion
- Integrated Circuits and Power Electronics
- Nanoelectronic Devices and Nanosystems
- Photonics, Nanoscience and Quantum Technology
- NEMS/MEMS

Interdisciplinary Research

EE faculty collaborate with researchers from other departments and schools across campus. More than a quarter of our faculty are joint with other departments, and a similar fraction of our PhD students have advisors outside EE. While the main applications of electrical engineering in the past four decades have been in information technology, EE tools and techniques are being increasingly applied more broadly to address major societal problems in areas such as:

Biomedical

Research in the biomedical area utilizes engineering approaches to address the unmet needs in diagnosis, staging, treatment, and mitigation of illnesses including cancer, diabetes, heart diseases, as well as brain disorders. Lower-cost, prevention-oriented health care delivery is critically needed, as well as new approaches to previously untreatable health conditions. Addressing these challenges requires discovering and creating

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fundamentally new approaches and creating new devices and systems for critical diagnostics (sensors, imaging), therapeutics (lasers, pacemakers, and neural interfaces), and analytical (high-throughput sequencing, healthcare IT) technologies.

Energy

Research in energy is motivated at the macro level by the rapid rise in worldwide demand for electricity and the threat of global climate change, and on the micro level by the explosion in the number of mobile devices and sensors whose performance and lifetimes are limited by energy.

On the macro level, electronic loads, such as data centers, smart appliances, and electric vehicles, are poised to overtake traditional industrial loads in consumption share. Renewable energy will make up at least half of the generation mix and drive adoption of novel technologies such as storage, fuel cells, waste-to-power and distributed generation. Our research investigates techniques such as demand response and the use of energy storage to reduce peak demand and address variability of renewable energy.

On the micro level, we are exploring energy efficient devices, power electronics, system architectures, and network protocols, as well as ways to harvest energy from the environment for wearable devices and the Internet of things.

For additional information, see the Department of Electrical Engineering's Research website.

Electrical Engineering Course Catalog Numbering System

Electrical Engineering courses are typically numbered according to the year in which the courses are normally taken.

	Electrical Engineering Course Catalog Numbering System
NUMBER	YEAR
010-099	first or second year undergraduate
100-199	second through fourth year undergraduate
200-299	mezzanine courses for advanced undergraduate or first-year graduate
300-399	second through fourth year graduate
400-499	specialized courses for advanced graduate
600-799	special summer courses

Faculty

Emeriti: (Professors) Clayton W. Bates, John Cioffi*, Donald C. Cox, Robert W. Dutton, Michael J. Flynn, James F. Gibbons, Andrea G. Goldsmith, Joseph W. Goodman, Robert M. Gray, James Harris, Stephen E. Harris, Martin E. Hellman, Umrhan S. Inan*, Thomas Kailath*, Gregory T.A. Kovacs*, Marc Levoy, Albert Macovski, Malcolm M. McWhorter, Teresa Meng, R. Fabian W. Pease, Leonard Tyler, Robert L. White, Bernard Widrow, Bruce A. Wooley, Yoshihisa Yamamoto; (Associate Professors) John T. Gill III, Bruce B. Lusignan; (Professors, Research) Antony Fraser-Smith*, C. Robert Helms, Leonid Kazovsky, Butrus Khuri-Yakub*, Ingolf Lindau*, David Luckham, Yoshio Nishi, Arogyaswami J. Paulraj

(*Recalled to active duty)

Chair: Jelena Vuckovic

Associate Chairs: John Pauly (*Undergraduate Education*), Brad Osgood (*Graduate Education*), Howard Zebker (*Admissions*), Kunle Olukotun (*Diversity and Inclusion*)

Academic Affairs Committee Chair: Joseph M. Kahn

Director of Graduate Studies: Brad Osgood

Professors: Nicholas Bambos, Kwabena Boahen, Dan Boneh, Stephen P. Boyd, Abbas El Gamal, Shanhui Fan, Bernd Girod, Patrick Hanrahan, John L. Hennessy, Lambertus Hesselink, Mark A. Horowitz, Roger T. Howe, Joseph M. Kahn, Christoforos E. Kozyrakis, Sanjay Lall, Thomas H. Lee, Nick McKeown, David A. B. Miller, Subhasish Mitra, Andrea Montanari, Boris Murmann, Dwight G. Nishimura, Oyekunle Olukotun, Brad G. Osgood, John M. Pauly, James D. Plummer, Eric Pop, Balaji Prabhakar, Mendel Rosenblum, Krishna Saraswat, Krishna V. Shenoy, H. Tom Soh, Olav Solgaard, Fouad A. Tobagi, David Tse, Benjamin Van Roy, Jelena Vuckovic, Shan X. Wang, Tsachy Weissman, Jennifer Widom, H. S. Philip Wong, S. Simon Wong, Howard Zebker

Associate Professors: Amin Arbabian, Srabanti Chowdhury, John Duchi, Dawson Engler, Jonathan Fan, Sachin Katti, Philip Levis, Ayfer Ozgur Aydin, Ada Poon, Juan Rivas-Davila, Gordon Wetzstein*

Assistant Professors: Sara Achour, Joonhee Choi, Daniel Congreve, Chelsea Finn, Mert Pilanci, Priyanka Raina, Dorsa Sadigh, Caroline Trippel, Mary Wootters

Professors (Research): Piero Pianetta

Courtesy Professors: Maneesh Agrawala, Stacey Bent, Kim Butts-Pauly, Emmanuel Candes, E.J. Chichilnisky, Amir Dembo, Utkan Demirci, Gary Glover, Peter Glynn, Leonidas Guibas, Brian Hargreaves, Tony Heinz, Ramesh Johari, Oussama Khatib, Monica S. Lam, Craig Levin, John C. Mitchell, Sandy Nepal, John Ousterhout, Daniel Palanker, Julius Smith, Dan Spielman, Brian Wandell, Lei Xing, Yinyu Ye

Courtesy Associate Professors: Mohsen Bayati, Sigrid Close, Todd Coleman, Adam de la Zerda, Surya Ganguli, Hanlee Ji, Jin Hyung Lee, Marco Pavone, Ram Rajagopal, Debbie Senesky, Kavin Setsompop, Kuang Xu

Courtesy Assistant Professors: Grace Gao, Scott Linderman, Paul Nuyujukian, Simona Onori, Adam Wang, Keith Winstein, Serena Yeung, Matei Zaharia, James Zou

Lecturers: Zain Asgar, Raul Camposano, Steven Clark, Andrea Di Blas, Antun Domic, Abbas Emami-Naeini, Joyce Farrell, Leslie Field, J., Patrick Groeneveld, My T. Le, Max Yuen

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Adjunct Professors: Sherif Ahmed, Ahmad Bahai, Rick Bahr, Fred M. Gibbons, Dmitry Gorinevsky, Bob S. Hu, Waguih Ishak, Theodore Kamins, Ali Keshavarzi, David Leeson, Liam Madden, Georgios Michelogiannakis, Fernando Mujica, Narasimha Madihally, Reza Nasiri Mahalati, Dan O'Neill, John Provine, Stephen Ryu, Ronald Schafer, Ashok Srivastava, David Sussillo, James Weaver

Visiting Professor: Zhishen Cheng, Peter Stoica

Visiting Associate Professor: Waleed Alhanafy, Jianing Li, Seunghyun Lee, Severin T. Schneebeli

Visiting Assistant Professor: Jun-Chau Chien

Programs

EE-BS - Electrical Engineering (BS)

Degree Designation: BS - Bachelor of Science

EE-MS - Electrical Engineering (MS)

Degree Designation: MS - Master of Science

EE-PMN - Electrical Engineering (PhD Minor)

Degree Designation:

EE-MIN - Electrical Engineering (Minor)

Degree Designation:

EE-PHD - Electrical Engineering (PhD)

Degree Designation: PHD - Doctor of Philosophy

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Courses

EE100 - ELEC ENGR PROFESSION

Credits: 1

Lectures/discussions on topics of importance to the electrical engineering professional. Continuing education, professional societies, intellectual property and patents, ethics, entrepreneurial engineering, and engineering management.

EE101B - CIRCUITS II

Credits: 4

Continuation of EE101A. Introduction to circuit design for modern electronic systems. Modeling and analysis of analog gain stages, frequency response, feedback. Filtering and analog to digital conversion. Fundamentals of circuit simulation. Prerequisites: EE101A, EE102A. Recommended: MATH 53 or CME102.

EE102B - SIGNAL PROCESS, LINEAR SYS II

Credits: 4

Continuation of EE 102A. Concepts and tools for continuous- and discrete-time signal and system analysis with applications in communications, signal processing and control. Analog and digital modulation and demodulation. Sampling, reconstruction, decimation and interpolation. Finite impulse response filter design. Discrete Fourier transforms, applications in convolution and spectral analysis. Laplace transforms, applications in circuits and feedback control. Z transforms, applications in infinite impulse response filter design. Prerequisite: EE 102A.

EE101A - CIRCUITS I

Credits: 4

Introduction to circuit modeling and analysis. Topics include creating the models of typical components in electronic circuits and simplifying non-linear models for restricted ranges of operation (small signal model); and using network theory to solve linear and non-linear circuits under static and dynamic operations. Prerequisite: ENGR40 or ENGR40M is strongly recommended.

EE102A - SIGNAL PROCESS, LINEAR SYS I

Credits: 4

Concepts and tools for continuous- and discrete-time signal and system analysis with applications in signal processing, communications, and control. Mathematical representation of signals and systems. Linearity and time invariance. System impulse and step responses. System frequency response. Frequency-domain representations: Fourier series and Fourier transforms. Filtering and signal distortion. Time/frequency sampling and interpolation. Continuous-discrete-time signal conversion and quantization. Discrete-time signal processing. Prerequisite: MATH 53 or CME 102.

EE104 - MACHINE LEARNING

Credits: 5

Introduction to machine learning. Formulation of supervised and unsupervised learning problems. Regression and classification. Data standardization and feature engineering. Loss function selection and its effect on learning. Regularization and its role in controlling complexity. Validation and overfitting. Robustness to outliers. Simple numerical implementation. Experiments on data from a wide variety of engineering and other disciplines. Undergraduate students should enroll for 5 units, and graduate students should enroll for 3 units. Prerequisites: ENGR 108; EE 178 or CS 109; CS106A or equivalent.

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EE107 - EMBEDDED NETWORKED SYSTEMS

Credits: 3

Networked embedded systems are often hidden from our view, but they are a key component that enables our modern society. Embedded systems bridge our physical world with powerful digital measurement and control systems. Applications of today's embedded systems range from stabilization in drones authentication in credit cards, and even temperature control in toasters. In this class, students will learn about how to build an networked embedded system from the ground up. The lectures will focus on the key enabling components of embedded systems, including: Clocks, GPIO, Interrupts, Busses, Amplifiers, Regulators, Power supplies, ADC/DAC, DMA, and Storage. The goal of the class is to familiarize the students with these components such that they can build their own embedded systems in devices. Prerequisites: EE 102A or ENGR 40M.

EE109 - DIGITAL SYSTEM DESIGN LAB

Credits: 4

The design of integrated digital systems encompassing both customized software and hardware. Software/hardware design tradeoffs. Algorithm design for pipelining and parallelism. System latency and throughput tradeoffs. FPGA optimization techniques. Integration with external systems and smart devices. Firmware configuration and embedded system considerations. Enrollment limited to 25; preference to graduating seniors. Prerequisites: 108B, and CS 106B or X.

EE116 - SEMICONDUCTOR DEVICES

Credits: 3

The underpinnings of modern technology are the transistor (circuits), the capacitor (memory), and the solar cell (energy). EE 116 introduces the physics of their operation, their historical origins (including Nobel prize breakthroughs), and how they can be optimized for future applications. The class covers physical principles of semiconductors, including silicon and new material discoveries, quantum effects, band theory, operating principles, and device equations. Recommended (but not required) co-requisite: EE 65 or equivalent.

EE11SC - DREAM IT, BUILD IT!

Credits: 2

The world is filled with electronic devices! There seem to be more and more all the time. Wouldn't it be cool to hack and build stuff? Bend electronics to your will? Cloud connect your own stuff? Dream It, Build It is a great place to start. Designed for folks with no experience, it will take you from zero to capable in short order. We will show you some of the worst kept secrets of how things are built and help you build stuff of your own. We'll start out with some basics about how to build things, how to measure things, how to hook stuff together and end up being able to make cloud-connected gizmos. [This is a SOPHOMORE COLLEGE course. Visit soco.stanford.edu for full details.]

EE12Q - SCIENCE, TECHNOLOGY, ART

Credits: 3

This course presents the interwoven histories of science, technology, and art starting in the late Medieval period in Europe, through the Renaissance, up to the Modern era. It explores how advances in science and technology were exploited by artists and how problems confronted by artists were often solved by scientists and technologists, to the advancement of all. Topics include the geometry of perspective, optics of image making, chemistry of pigments and dyes, and the role of computing in art. A subsidiary theme is how artists indirectly interpreted scientific discoveries (telescope views of the heavens, microscope views of the teeny, Theory of Relativity, ...). Whenever possible, the technical evidence, developments, and of course art will be presented visually in the class.

EE108 - DIGITAL SYSTEM DESIGN

Credits: 5

Digital circuit, logic, and system design. Digital representation of information. CMOS logic circuits. Combinational logic design. Logic building blocks, idioms, and structured design. Sequential logic design and timing analysis. Clocks and synchronization. Finite state machines. Microcode control. Digital system design. Control and datapath partitioning. Lab. *In Autumn, enrollment preference is given to EE majors. Any EE majors who must enroll in Autumn are invited to contact the instructor. Formerly EE 108A.

EE114 - FUND OF ANLG INTEGRTD CIRC DES

Credits: 4

Analysis and simulation of elementary transistor stages, current mirrors, supply- and temperature-independent bias, and reference circuits. Overview of integrated circuit technologies, circuit components, component variations and practical design paradigms. Differential circuits, frequency response, and feedback will also be covered. Performance evaluation using computer-aided design tools. Undergraduates must take EE 114 for 4 units. Prerequisite: 101B. GER:DB-EngrAppSci

EE118 - INTRODUCTION TO MECHATRONICS

Credits: 4

Technologies involved in mechatronics (intelligent electro-mechanical systems), and techniques to apply this technology to mecatronic system design. Topics include: electronics (A/D, D/A converters, op-amps, filters, power devices); software program design, event-driven programming; hardware and DC stepper motors, solenoids, and robust sensing. Large, open-ended team project. Prerequisites: ENGR 40, CS 106, or equivalents.

EE124 - INTRO TO NUEROELECTRICAL ENGR

Credits: 3

Fundamental properties of electrical activity in neurons, technology for measuring and altering neural activity, and operating principles of modern neurological and neural prosthetic medical systems. Topics: action potential generation and propagation, neuro-MEMS and measurement systems, experimental design and statistical data analysis, information encoding and decoding, clinical diagnostic systems, and fully-implantable neural prosthetic systems design. Prerequisite: EE 101A and EE 102A.

EE133 - ANALOG COMMUN DESIGN LAB

Credits: 4

Design, testing, and applications of Radio Frequency (RF) electronics: Amplitude Modulation (AM), Frequency Modulation (FM) and concepts of Software Define Radio (SDR) systems. Practical aspects of circuit implementations are developed; labs involve building and characterization of subsystems as well as integration of a complete radio system and a final project. Total enrollment limited to 25 students, undergraduate and graduate levels. Prerequisite: EE101B. Undergraduate students enroll in EE133 for 4 units and Graduate students enroll in EE233 for 3 units. Recommended: EE114/214A.

EE134 - INTRODUCTION TO PHOTONICS

Credits: 4

Optics and photonics underpin the technologies that define our daily life, from communications and sensing to displays and imaging. This course provides an introduction to the principles that govern the generation, manipulation, and detection of light and will give students hands-on lab experience applying these principles to analyze and design working optical systems. The concepts we will cover form the basis for many systems in biology, optoelectronics, and telecommunications and build a foundation for further learning in photonics and optoelectronics. Connecting theory to observation and application is a major theme for the course. Prerequisite: EE 102A and one of the following: EE 42, Physics 43, or Physics 63.

EE14N - THINGS ABOUT STUFF

Credits: 3

Preference to freshmen. The stories behind disruptive inventions such as the telegraph, telephone, wireless, television, transistor, and chip are as important as the inventions themselves, for they elucidate broadly applicable scientific principles. Focus is on studying consumer devices; projects include building batteries, energy conversion devices and semiconductors from pocket change. Students may propose topics and projects of interest to them. The trajectory of the course is determined in large part by the students themselves.

EE155 - GREEN ELECTRONICS

Credits: 4

Many green technologies including hybrid cars, photovoltaic energy systems, efficient power supplies, and energy-conserving control systems have at their heart intelligent, high-power electronics. This course examines this technology and uses green-tech examples to teach the engineering principles of modeling, optimization, analysis, simulation, and design. Topics include power converter topologies, periodic steady-state analysis, control, motors and drives, photovoltaic systems, and design of magnetic components. The course involves a hands-on laboratory and a substantial final project. Formerly EE 152. Required: EE101B, EE102A, EE108. Recommended: ENGR40 or EE122A.

EE142 - ENGINEERING ELECTROMAGNETICS

Credits: 3

Introduction to electromagnetism and Maxwell's equations in static and dynamic regimes. Electrostatics and magnetostatics: Gauss's, Coulomb's, Faraday's, Ampere's, Biot-Savart's laws. Electric and magnetic potentials. Boundary conditions. Electric and magnetic field energy. Electrodynamics: Wave equation; Electromagnetic waves; Phasor form of Maxwell's equations. Solution of the wave equation in 1D free space: Wavelength, wave-vector, forward and backward propagating plane waves. Poynting's theorem. Propagation in lossy media, skin depth. Reflection and refraction at planar boundaries, total internal reflection. Solutions of wave equation for various 1D-3D problems: Electromagnetic resonators, waveguides periodic media, transmission lines. Formerly EE 141. Prerequisites: an introductory course in electromagnetics (PHYSICS 43, PHYSICS 63, PHYSICS 81, or EE 42) and a solid background in vector calculus (CME 100, CME 102, or MATH 52, with MATH 52 being an ideal prerequisite)

EE153 - POWER ELECTRONICS

Credits: 4

Addressing the energy challenges of today and the environmental challenges of the future will require efficient energy conversion techniques. This course will discuss the circuits used to efficiently convert ac power to dc power, dc power from one voltage level to another, and dc power to ac power. The components used in these circuits (e.g., diodes, transistors, capacitors, inductors) will also be covered in detail to highlight their behavior in a practical implementation. A lab will be held with the class where students will obtain hands on experience with power electronic circuits. For WIM credit, students must enroll in EE 153 for 4 units. No exceptions. Formerly EE 292J. Prerequisite: EE 101A. Strongly recommended EE 101B.

EE156 - BOARD LEVEL DESIGN

Credits: 4

The ability to rapidly create board level electronics at prototype and short run volumes is enabling; Board Level Design teaches how to do this. This course focuses on applying circuit design concepts to rapidly create electronics to augment existing research instruments, explore and reduce technical risk, and provide engineering samples for evaluation. Students will send several PCBs for fabrication during the Quarter. The PCBs will be "brought-up" and tested to confirm functionality and performance to specification. Undergraduate EE majors will gain deeper exposure to circuits and planar electromagnetics. Students must enroll in EE 156 for 4 units and EE 256 for 3 units. Prerequisites: EE 42, EE 101A, and EE 108 or consent of instructor.

EE157 - APPLICATIONS OF ELECTRIC MOTOR

Credits: 3

An introduction to electric motors and the principles of electromechanical energy conversion. Students will learn about, design, and build an electric motor system, choosing from one of three application areas: renewable energy (wind turbines), robotics (drones and precision manufacturing), or electric vehicles (cars, ships, and airplanes). Topics covered include ac and dc rotating machines, power electronics inverters and drives, and control techniques. Prerequisite: EE 42, Physics 43, ENGR 40M or equivalent.

EE160A - PRINCIPLES OF ROBOT AUTONOMY I

Credits: 4

Basic principles for endowing mobile autonomous robots with perception, planning, and decision-making capabilities. Algorithmic approaches for robot perception, localization, and simultaneous localization and mapping; control of non-linear systems, learning-based control, and robot motion planning; introduction to methodologies for reasoning under uncertainty, e.g., (partially observable) Markov decision processes. Extensive use of the Robot Operating System (ROS) for demonstrations and hands-on activities. Prerequisites: CS 106A or equivalent, CME 100 or equivalent (for linear algebra), and CME 106 or equivalent (for probability theory).

EE169 - INTRO TO BIOIMAGING

Credits: 3

Bioimaging is important for both clinical medicine, and medical research. This course will provide a introduction to several of the major imaging modalities, using a signal processing perspective. The course will start with an introduction to multi-dimensional Fourier transforms, and image quality metrics. It will then study projection imaging systems (projection X-Ray), backprojection based systems (CT, PET, and SPECT), systems that use beam forming (ultrasound), and systems that use Fourier encoding (MRI). Prerequisites: EE102A, EE102B

EE15Q - ART & SCIENCE ENGR DESIGN

Credits: 3

The goal of this seminar is to introduce sophomores to the design process associated with an engineering project. The seminar will consist of a series of lectures. The first part of each lecture will focus on the different design aspects of an engineering project, including formation of the design team, developing a project statement, generating design ideas and specifications, finalizing the design, and reporting the outcome. Students will form teams to follow these procedures in designing a term project of their choice over the quarter. The second part of each lecture will consist of outside speakers, including founders of some of the most exciting companies in Silicon Valley, who will share their experiences about engineering design. On-site visits to Silicon Valley companies to showcase their design processes will also be part of the course. The seminar serves three purposes: (1) it introduces students to the design process of turning an idea into a final design, (2) it presents the different functions that people play in a project, and (3) it gives students a chance to consider what role in a project would be best suited to their interests and skills.

EE168 - INTRO DIGITAL IMAGE PROCESSING

Credits: 4

Computer processing of digital 2-D and 3-D data, combining theoretical material with implementation of computer algorithms. Topics: properties of digital images, design of display systems and algorithms, time and frequency representations, filters, image formation and enhancement, imaging systems, perspective, morphing, and animation applications. Instructional computer lab exercises implement practical algorithms. Final project consists of computer animations incorporating techniques learned in class. For WIM credit, students must enroll for 4 units. No exceptions. Prerequisite: Matlab programming.

EE178 - INTR PROBABLISTIC SYS ANALYSIS

Credits: 4

Introduction to probability and its role in modeling and analyzing real world phenomena and systems, including topics in statistics, machine learning, and statistical signal processing. Elements of probability, conditional probability, Bayes rule, independence. Discrete and continuous random variables. Signal detection. Functions of random variables. Expectation; mean, variance and covariance, linear MSE estimation. Conditional expectation; iterated expectation, MSE estimation, quantization and clustering. Parameter estimation. Classification. Sample averages. Inequalities and limit theorems. Confidence intervals. Prerequisites: Calculus at the level of MATH 51, CME 100 or equivalent and basic knowledge of computing at the level of CS106A.

EE179 - ANALOG AND DIGITAL COMMUNICATIONS

Credits: 3

This course covers the fundamental principles underlying the analysis, design and optimization of analog and digital communication systems. Design examples will be taken from the most prevalent communication systems today: cell phones, Wifi, radio and TV broadcasting, satellites, and computer networks. Analysis techniques based on Fourier transforms and energy/power spectral density will be developed. Mathematical models for random variables and random (noise) signals will be presented, which are used to characterize filtering and modulation of random noise. These techniques will then be used to design analog (AM and FM) and digital (PSK and FSK) communication systems and determine their performance over channels with noise and interference. Prerequisite: 102A.

EE180 - DIGITAL SYSTEMS ARCHITECTURE

Credits: 4

The design of processor-based digital systems. Instruction sets, addressing modes, data types. Assembly language programming, low-level data structures, introduction to operating systems and compilers. Processor microarchitecture, microprogramming, pipelining. Memory systems and caches. Input/output, interrupts, buses and DMA. System design implementation alternatives, software/hardware tradeoffs. Labs involve the design of processor subsystems and processor-based embedded systems. Formerly EE 108B. Prerequisite: one of CS107 or CS 107E (required) and EE108 (recommended but not required).

EE185 - INTERACTIVE LIGHT SCULPTURE

Credits: 9

Design, prototype, build, refine, program, and install a large interactive light sculpture in the Packard Building to celebrate the 125th anniversary of the EE department. Students may take the course for 1, 2, or 3 quarters; each quarter focuses on a different phase of the project. Topics covered include energy budgeting, communication, enclosure design, scalability, timing, circuit design, structural design, and safety. Prerequisite: ENGR 40M, or an introductory EE or CS course in circuits or programming.

EE185B - ENGINEERING A SMART OBJECT

Credits: 3

EE 185A/B/C is a full-year sequence that teaches all of the concepts, knowledge, skills, and techniques to engineer all aspects of a smart object. This second course focuses on understanding the art of specification by writing a specification and fabricating to someone else is written specification. We will also explore embedded system design and the impact of design decisions by redesigning the electronics from EE 185A to meet low power specifications. Students will learn about power, energy, micro controllers, low-level software and how, in embedded systems, electronic hardware, mechanical design, and software are coupled. Course prerequisites: EE 185A as well as CS107, CS107E or instructor approval.

EE17N - ENGR MICRO & NANO WORLDS

Credits: 3

Preference to freshmen. The first part is hands-on micro- and nano-fabrication including the Stanford Nanofabrication Facility (SNF) and the Stanford Nanocharacterization Laboratory (SNL) and field trips to local companies and other research centers to illustrate the many applications; these include semiconductor integrated circuits ('chips'), DNA microarrays, microfluidic bio-sensors and microelectromechanical systems (MEMS). The second part is to create, design, propose and execute a project. Most of the grade will be based on the project. By the end of the course you will, of course, be able to read critically a New York Times article on nanotechnology. More importantly you will have experienced the challenge (and fun) of designing, carrying out and presenting your own experimental project. As a result you will be better equipped to choose your major. This course can complement (and differs from) the seminars offered by Profs Philip Wong and Hari Manoharan in that it emphasizes laboratory work and an experimental student-designed project. Prerequisites: high-school physics.

EE184 - INTERNET PRINCIPLES AND PROTOCOLS

Credits: 3

This course covers the basic functions underlying computer networks and their organization into a layered architecture. The principles set forth for internetworking that allowed the Internet to be open and scalable are highlighted. Addressing in the Internet, the Internet Protocol (IP), the Transmission Control Protocol (TCP), and the various routing protocols used in the Internet are examined. The course also examines the design of specific prevalent networks (Ethernet and Wi-Fi, in particular) addressing both Physical Layer functionality (how bits are signaled on the transmission medium) and Media Access Control (MAC) Layer functionality which comprises how stations are addressed, the protocol according to which stations access a common shared transmission medium, and various management functions necessary for the operation of the network).

EE185A - ENGINEERING A SMART OBJECT

Credits: 3

EE 185A/B/C is a full-year sequence that teaches all of the concepts, knowledge, skills, and techniques to engineer all aspects of a smart object. Students learn to specify and analyze designs precisely, such that the first version of the object constructed works exactly as expected. This first course focuses on building an object to specification. Students will learn and use modern fabrication methods including machine tools (milling machine, lathe, etc.), laser cutters, and 3D printers to demonstrate their ability to build it right the first time. Course prerequisites: Physics 43 (or equivalent) and ENGR 40M or instructor approval.

EE185C - ENGINEERING A SMART OBJECT

Credits: 3

EE 185A/B/C is a full-year sequence that teaches all of the concepts, knowledge, skills, and techniques to engineer all aspects of a smart object. In this third course, the students bring everything they have learned in EE 185 A/B to bear by engineering a simple smart object of their choosing. We will add an essential ingredient of a Smart Object - connectivity and learn about how this effects system design. During this Quarter, each student will write a precise specification, create and analyze their design to a degree such that it is certain it will work as intended the first time they build it. They will also fabricate and demonstrate their Smart Object by the end of the Quarter. Course prerequisites: EE 185B or instructor approval.

EE190 - SPECIAL STUDIES OR PROJ IN EE

Credits: 999

Independent work under the direction of a faculty member. Individual or team activities involve lab experimentation, design of devices or systems, or directed reading. Course may be repeated for credit.

EE191A - SPECIAL STUDIES AND REPORTS

Credits: 1

EE191A is part of the Accelerated Calculus for Engineers program. Independent work under the direction of a faculty member given for a letter grade only. EE 191A counts as a Math one unit seminar course: it is this unit that constitutes the ACE program.

EE192T - THEATRE TECH IN AGE OF COVID

Credits: 3

This class is part of a multi-disciplinary collaboration between researchers in the CS, EE, and TAPS departments to design and develop a system to host a live theatrical production that will take place over the Internet in the winter quarter. The performing arts have been greatly affected by a transition to theater over Zoom and its competitors, none of which are great at delivering low-latency audio to actors, or high-quality audio and video to the audience, or feedback from the audience back to actors. These are big technical challenges. During the fall, we'll build a system that improves on current systems in certain areas: audio quality and latency over spotty Internet connections, video quality and realistic composited scenes with multiple actors, audience feedback, and perhaps digital puppetry. Students will learn to be part of a deadline-driven software development effort working to meet the needs of a theater director and creative specialists -- while communicating the effect of resource limits and constraints to a nontechnical audience. This is an experimental hands-on laboratory class, and our direction may shift as the creative needs of the theatrical production evolve. Based on the success of class projects and subsequent needs, some students may be invited to continue in the winter term with a research appointment (for pay or credit) to operate the system you have built and instruct actors and creative professionals how to work with the system through rehearsals and the final performance before spring break. Prerequisites: CS110 or EE102A. Recommended: familiarity with Linux, C++, and Git.

EE203 - THE ENTREPRENEURIAL ENGINEER

Credits: 999

Seminar. For prospective entrepreneurs with an engineering background. Contributions made to the business world by engineering graduates. Speakers include Stanford and other engineering and M.B.A. graduates who have founded large and small companies in nearby communities. Contributions from EE faculty and other departments including Law, Business, and MS&E. May be repeated for credit.

EE191 - SPECIAL STUDIES/REPORTS IN EE

Credits: 999

Independent work under the direction of a faculty member given for a letter grade only. If a letter grade given on the basis of required written report or examination is not appropriate, enroll in 190. Course may be repeated for credit.

EE191W - SPECIAL STUDIES & REPORTS WIM

Credits: 10

WIM-version of EE 191. For EE students using special studies (e.g., honors project, independent research project) to satisfy the writing-in-major requirement. A written report that has gone through revision with an adviser is required. An adviser from the Technical Communication Program is recommended.

EE195 - EE INSTRUCTION

Credits: 3

Students receive training from faculty or graduate student mentors to prepare them to assist in instruction of Electrical Engineering courses. The specific training and units of credit received are to be defined in consultation with one of the official instructors of EE 195. Note that University regulations prohibit students from being paid for the training while receiving academic credit for it. Enrollment limited.

EE205 - PRODUCT MANAGEMENT

Credits: 3

Successful products are the highest impact contribution anyone can make in product development. Students will learn to build successful products using fundamental concepts in Product Management. These include understanding customers, their job to be done, identifying new product opportunities, and defining what to build that is technically feasible, valuable to the customer, and easy to use. The course has two components, Product Management Project with corporate partners, and case-based classroom discussion of PM concepts and application. Prerequisite: Students must be currently enrolled in a MS or PhD engineering degree program.

EE207 - NEUROMORPHICS: BRAINS IN SILIC

Credits: 3

(Formerly EE 304) Neuromorphic systems run perceptual, cognitive and motor tasks in real-time on a network of highly interconnected nonlinear units. To maximize density and minimize energy, these units--like the brain's neurons--are heterogeneous and stochastic. The first half of the course covers learning algorithms that automatically synthesize network configurations to perform a desired computation on a given heterogeneous neural substrate. The second half of the course surveys system-on-a-chip architectures that efficiently realize highly interconnected networks and mixed analog-digital circuit designs that implement area and energy-efficient nonlinear units. Prerequisites: EE102A is required.

EE214A - FUND OF ANLG INTEGRTD CIRC DES

Credits: 4

Analysis and simulation of elementary transistor stages, current mirrors, supply- and temperature-independent bias, and reference circuits. Overview of integrated circuit technologies, circuit components, component variations and practical design paradigms. Differential circuits, frequency response, and feedback will also be covered. Performance evaluation using computer-aided design tools. Undergraduates must take EE 114 for 4 units. Prerequisite: 101B. GER:DB-EngrAppSci

EE216 - PRIN/MODELS SEMICOND DEVICES

Credits: 3

Carrier generation, transport, recombination, and storage in semiconductors. Physical principles of operation of the p-n junction, heterojunction, metal semiconductor contact, bipolar junction transistor, MOS capacitor, MOS and junction field-effect transistors, and related optoelectronic devices such as CCDs, solar cells, LEDs, and detectors. First-order device models that reflect physical principles and are useful for integrated-circuit analysis and design. Prerequisite: 116 or equivalent.

EE21N - MAKING AT NANOMETER SCALE

Credits: 3

Have you ever wondered what is inside your phone and your computer? What physical events happen in between the time you press the 'search' button and the information shows up on the screen? In this course, we start with the classic paper by Richard Feynman, "There's Plenty of Room at the Bottom," which laid down a challenge to the nanotechnologists. Today's microchips are nanotechnology in action. Transistors are nanometer scale. We will introduce students to the tools of nanotechnologists and the basic elements of nanoscale science and engineering such as nanotubes, nanowires, nanoparticles, and self-assembly. We will visit nanotechnology laboratories to consolidate our learning, go into the Stanford Nanofabrication Facility (SNF), and do a four-week project on nanofabrication. Hands-on laboratory work will be introduced (e.g., lithography, seeing things at the nanoscale using electron microscopes). We will learn how to build transistors from scratch and test them.

EE212 - INTEG CIRCUIT FABRICATION PROC

Credits: 3

For students interested in the physical bases and practical methods of silicon VLSI chip fabrication, or the impact of technology on device and circuit design, or intending to pursue doctoral research involving the use of Stanford's Nanofabrication laboratory. Process simulators illustrate concepts. Topics: principles of integrated circuit fabrication processes, physical and chemical models for crystal growth, oxidation, ion implantation, etching, deposition, lithography, and back-end processing. Required for 410.

EE214B - ADV INTEGRATED CIRC DESIGN

Credits: 3

Analysis and design of analog and digital integrated circuits in advanced CMOS technology. Emphasis on compact modeling of performance limiting aspects and intuitive approaches to design. Analytical treatment of noise; analog circuit sizing using the transconductance to current ratio; analysis and design of feedback circuits. Delay analysis of digital logic gates; decoder design using logical effort. CMOS image sensors are used as a motivating application example. Prerequisites: EE114/214A.

EE218 - POWER SEMICONDUCTOR

Credits: 3

This course starts by covering the device physics and technology of current silicon power semiconductor devices including power MOSFETs, IGBTs, and Thyristors. Wide bandgap materials, especially GaN and SiC are potential replacements for Si power devices because of their fundamentally better properties. This course explores what is possible in these new materials, and what the remaining challenges are for wide bandgap materials to find widespread market acceptance in power applications. Future clean, renewable energy systems and high efficiency power control systems will critically depend on the higher performance devices possible in these new materials. Prerequisites: EE 116 or equivalent.

EE222 - EMPAPPLIED QUANTUM MECHANICS I

Credits: 3

Emphasis is on applications in modern devices and systems. Topics include: Schrödinger's equation, eigenfunctions and eigenvalues, solutions of simple problems including quantum wells and tunneling, quantum harmonic oscillator, coherent states, operator approach to quantum mechanics, Dirac notation, angular momentum, hydrogen atom, calculation techniques including matrix diagonalization, perturbation theory, variational method, and time-dependent perturbation theory with applications to optical absorption, nonlinear optical coefficients, and Fermi's golden rule. Prerequisites: MATH 52 and 53, one of EE 65, ENGR 65, PHYSICS 71 (formerly PHYSICS 65), PHYSICS 70.

EE223 - APPLIED QUANTUM MECHANICS II

Credits: 3

Continuation of 222, including more advanced topics: quantum mechanics of crystalline materials, methods for one-dimensional problems, spin, systems of identical particles (bosons and fermions), introductory quantum optics (electromagnetic field quantization, coherent states), fermion annihilation and creation operators, interaction of different kinds of particles (spontaneous emission, optical absorption, and stimulated emission). Quantum information and interpretation of quantum mechanics. Other topics in electronics, optoelectronics, optics, and quantum information science. Prerequisite: 222.

EE225 - BIOCHIPS AND MEDICAL IMAGING

Credits: 3

The course covers state-of-the-art and emerging bio-sensors, bio-chips, imaging modalities, and nano-therapies which will be studied in the context of human physiology including the nervous system, circulatory system and immune system. Medical diagnostics will be divided into bio-chips (in-vitro diagnostics) and medical and molecular imaging (in-vivo imaging). In-depth discussion on cancer and cardiovascular diseases and the role of diagnostics and nano-therapies.

EE233 - ANALOG COMMUN DESIGN LAB

Credits: 4

Design, testing, and applications of Radio Frequency (RF) electronics: Amplitude Modulation (AM), Frequency Modulation (FM) and concepts of Software Define Radio (SDR) systems. Practical aspects of circuit implementations are developed; labs involve building and characterization of subsystems as well as integration of a complete radio system and a final project. Total enrollment limited to 25 students, undergraduate and graduate levels. Prerequisite: EE101B. Undergraduate students enroll in EE133 for 4 units and Graduate students enroll in EE233 for 3 units. Recommended: EE114/214A.

EE235A - ANALYTICAL METHODS

Credits: 3

This course provides fundamental principles underlying important analytical techniques used in modern biotechnology. The course comprises of lectures and hands-on laboratory experiments. Students will learn the core principles for designing, implementing and analyzing central experimental methods including polymerase chain reaction (PCR), electrophoresis, immunoassays, and high-throughput sequencing. The overall goal of the course is to enable engineering students with little or no background in molecular biology to transition into research in the field of biomedicine.

EE236A - MODERN OPTICS

Credits: 3

Geometrical optics; lens analysis and design, aberrations, optical instruments, radiometry. ray matrices. Wave nature of light; polarization, plane waves at interfaces and in media with varying refractive index, diffraction, Fourier Optics, Gaussian beams. Interference; single-beam interferometers (Fabry-Perot), multiple-beam interferometers (Michelson, Mach-Zehnder). Prerequisites: EE 142 or familiarity with electromagnetism and plane waves.

EE224 - QUANTUM CONTROL

Credits: 3

Introduction to quantum control, dynamics, and information processing, aimed for graduate students and advanced undergraduate students. Prerequisites include some knowledge of elementary quantum mechanics, linear algebra, and statistical analysis. The course will provide an overview on both the fundamentals and state-of-the-art techniques in the area of quantum engineering. Topics include qubits and operators, modeling and numerical analysis of open quantum systems, quantum control protocols, average Hamiltonian theory, dynamical decoupling, quantum device benchmarking, different quantum platforms and their applications.

EE228 - BASIC PHYSOLID STATE ELECTRON

Credits: 3

Solid state devices have driven widespread technological revolution and are ubiquitous in our daily lives. We study the physics of solid state materials, enabling a complete understanding from the atom to the device. Topics include: energy band theory of solids; heterostructures and low-dimensional structures for bandgap engineering; electrons, holes, densities of states and relation to absorption and gain; and semiconductor statistics determining equilibrium and non-equilibrium carrier distributions. We explain how these principles govern the operation of modern devices, including transistors, light-emitting diodes and solar cells. Prerequisite: course in modern physics.

EE234 - PHOTONICS LABORATORY

Credits: 3

Photonics and fiber optics with a focus on communication and sensing. Experimental characterization of semiconductor lasers, optical fibers, photodetectors, receiver circuitry, fiber optic links, optical amplifiers, and optical sensors and photonic crystals. Prerequisite: EE 236A (recommended).

EE235B - ANALYTICAL METHODS IN BIOTECH

Credits: 3

This course is intended for graduate students, who are interested in biomedical research but have little background in fundamental laboratory techniques. Required prerequisite is EE235A/BIOS212/RAD236. This course seeks to equip such students with basic biochemistry and molecular biology techniques for them to pursue their research interests in biotechnology. The course will consist of a series of lectures and laboratory experiments.

EE236AL - MODERN OPTICS - LABORATORY

Credits: 1

The Laboratory Course allows students to work hands-on with optical equipment to conduct five experiments that compliment the lecture course. Examples are Gaussian Beams and Resonators, Interferometers, and Diffraction.

EE236B - GUIDED WAVES

Credits: 3

Maxwell's equations, constitutive relations. Kramers-Kronig relations. Modes in waveguides: slab, rectangular, circular. Photonic crystals, surface plasmon modes. General properties of waveguide modes: orthogonality, phase and group indices, group velocity dispersion. Chirped pulse propagation in dispersive media and its connection to Gaussian beam propagation. Time lens. Waveguide technologies: glass, silicon, III-V semiconductor, metallic. Waveguide devices: fibers, lasers, modulators, arrayed waveguide gratings. Scattering matrix description of passive optical devices, and constraints from energy conservation, time-reversal symmetry and reciprocity. Mode coupling, directional couplers, distributed-feedback structures. Resonators from scattering matrix and input-output perspective. Micro-ring resonators. Prerequisites: EE 236A and EE 242 or familiarity with differential form of Maxwell's equations.

EE237 - SOLAR ENERGY CONVERSION

Credits: 3

This course will be an introduction to solar photovoltaics. No prior photovoltaics knowledge is required. Class lectures will be supplemented by guest lectures from distinguished engineers, entrepreneurs and venture capitalists actively engaged in solar industry. Past guest speakers include Richard Swanson (CEO, SunPower), Benjamin Cook (Managing Partner at NextPower Capital) and Shahin Farshchi (Partner, Lux Capital). Topics Include: Economics of solar energy. Solar energy policy. Solar cell device physics: electrical and optical. Different generations of photovoltaic technology: crystalline silicon, thin film, multi-junction solar cells. Perovskite and silicon tandem cells. Advanced energy conversion concepts like photon up-conversion, quantum dot solar cells. Solar system issues including module assembly, inverters, micro-inverters and microgrid. No prior photovoltaics knowledge is required. Recommended: EE116, EE216 or equivalent.

EE242 - ELECTROMAGNETIC WAVES

Credits: 3

This course will provide an advanced treatment of electromagnetic waves in free space and media. The first part of the course will cover reflection, refraction, resonators, photonic crystals, and waveguides. The second part will cover finite-difference time-domain (FDTD) computation and introduce students to commercial FDTD software. The third part will focus on an analysis of EM waves in matter. The fourth part will cover potentials, Green's functions, far-field radiation, near-field radiation, antennas, and phased arrays. In lieu of a final exam, students will perform a group project demonstrating theoretical and application proficiency in a topic of their choosing. Homeworks and the final project will tie into real world applications of electromagnetics and utilize scientific computing (Matlab, Mathematica, or Python). Prerequisites: EE 142 or PHYSICS 120, and prior programming experience (Matlab or other language at level of CS 106A or higher).

EE236C - LASERS

Credits: 3

Atomic systems, spontaneous emission, stimulated emission, amplification. Three- and four-level systems, rate equations, pumping schemes. Laser principles, conditions for steady-state oscillation. Transverse and longitudinal mode control and tuning. Exemplary laser systems: gas (HeNe), solid state (Nd:YAG, Ti:sapphire) and semiconductors. Elements of laser dynamics and noise. Formerly EE231. Prerequisites: EE 236B and familiarity with modern physics and semiconductor physics. Recommended: EE 216 and EE 223 (either may be taken concurrently).

EE238 - FOURIER OPTICS

Credits: 3

Fourier analysis applied to optical imaging. Theoretical topics include Fourier transform and angular spectrum to describe diffraction, Fourier transforming properties of lenses, image formation with coherent and incoherent light and aberrations. Application topics will cover image deconvolution/reconstruction, amplitude and phase pupil engineering, computational adaptive optics, and others motivated by student interest. Prerequisites: familiarity with Fourier transform and analysis, EE 102 and EE 142 or equivalent.

EE243 - SEMI-CONDUCTOR OPTOELECTRONIC

Credits: 3

Semiconductor physics and optical processes in semiconductors. Operating principles and practical device features of semiconductor optoelectronic materials and heterostructures. Devices include: optical detectors (p-i-n, avalanche, and MSM); light emitting diodes; electroabsorptive modulators (Franz-Keldysh and QCSE), electrorefractive (directional couplers, Mach-Zehnder), switches (SEEDs); and lasers (waveguide and vertical cavity surface emitting). Prerequisites: semiconductor devices and solid state physics such as EE 216 or equivalent.

EE247 - INTR OPTICAL FIBER COMM

Credits: 3

Fibers: single- and multi-mode, attenuation, modal dispersion, group-velocity dispersion, polarization-mode dispersion. Nonlinear effects in fibers: Raman, Brillouin, Kerr. Self- and cross-phase modulation, four-wave mixing. Sources: light-emitting diodes, laser diodes, transverse and longitudinal mode control, modulation, chirp, linewidth, intensity noise. Modulators: electro-optic, electro-absorption. Photodiodes: p-i-n, avalanche, responsivity, capacitance, transit time. Receivers: high-impedance, transimpedance, bandwidth, noise. Digital intensity modulation formats: non-return-to-zero, return-to-zero. Receiver performance: Q factor, bit-error ratio, sensitivity, quantum limit. Sensitivity degradations: extinction ratio, intensity noise, jitter, dispersion. Wavelength-division multiplexing. System architectures: local-area, access, metropolitan-area, long-haul. Prerequisites: EE 102A and EE 142.

EE252 - ANTENNAS

Credits: 3

This course aims to cover the theory, simulation, and hands-on experiment in antenna design. Topics include: basic parameters to describe the performance and characteristics of an antenna, link budget analyses, solving the fields from a Hertzian dipole, duality, equivalence principle, reciprocity, linear wire antenna, circular loop antenna, antenna array, slot and patch antennas, helical antennas, wideband antennas, size reduction techniques, wideband small antennas, and circularly polarized (CP) small antennas. Students will learn to use a commercial electromagnetic simulator in lab sessions. A final project is designed to solve a research antenna design problem in biomedical area or wireless communications. Prerequisite: EE 142 or Physics 120 or equivalent. Enrollment capacity limited to 25 students.

EE254 - ADVANCED TOPICS IN POWER ELECT

Credits: 3

In this course, we will study the practical issues related to the practical design of power electronic converters. We will also explore the trade-offs involved in selecting among the different circuits used to convert ac to dc, dc to ac and back to dc over a wide range of power levels suitable for different applications. In Advanced Topics in Power Electronic, as a multidisciplinary field, we will discuss power electronics circuits, extraction of transfer functions in Continuous and discontinuous conduction mode, voltage and current control of power converters, design of input/output filters to meet Electro Magnetic Interference specifications, layout of power electronics circuits and put this knowledge in a very practical context. Prerequisites: EE 153/253.

EE256 - BOARD LEVEL DESIGN

Credits: 4

The ability to rapidly create board level electronics at prototype and short run volumes is enabling; Board Level Design teaches how to do this. This course focuses on applying circuit design concepts to rapidly create electronics to augment existing research instruments, explore and reduce technical risk, and provide engineering samples for evaluation. Students will send several PCBs for fabrication during the Quarter. The PCBs will be "brought-up" and tested to confirm functionality and performance to specification. Undergraduate EE majors will gain deeper exposure to circuits and planar electromagnetics. Students must enroll in EE 156 for 4 units and EE 256 for 3 units. Prerequisites: EE 42, EE 101A, and EE 108 or consent of instructor.

EE251 - HIGH-FREQUENCY CIRCUIT DESIGN

Credits: 3

Students will study the theory of operation of instruments such as the time-domain reflectometer, sampling oscilloscope and vector network analyzer. They will build on that theoretical foundation by designing, constructing and characterizing numerous wireless building blocks in the upper-UHF range (e.g., up to about 500MHz), in a running series of laboratory exercises that conclude in a final project. Examples include impedance-matching and coupling structures, filters, narrowband and broadband amplifiers, mixers/modulators, and voltage-controlled oscillators. Prerequisite: EE 114 or EE 214A.

EE253 - POWER ELECTRONICS

Credits: 4

Addressing the energy challenges of today and the environmental challenges of the future will require efficient energy conversion techniques. This course will discuss the circuits used to efficiently convert ac power to dc power, dc power from one voltage level to another, and dc power to ac power. The components used in these circuits (e.g., diodes, transistors, capacitors, inductors) will also be covered in detail to highlight their behavior in a practical implementation. A lab will be held with the class where students will obtain hands on experience with power electronic circuits. For WIM credit, students must enroll in EE 153 for 4 units. No exceptions. Formerly EE 292J. Prerequisite: EE 101A. Strongly recommended EE 101B.

EE255 - GREEN ELECTRONICS

Credits: 4

Many green technologies including hybrid cars, photovoltaic energy systems, efficient power supplies, and energy-conserving control systems have at their heart intelligent, high-power electronics. This course examines this technology and uses green-tech examples to teach the engineering principles of modeling, optimization, analysis, simulation, and design. Topics include power converter topologies, periodic steady-state analysis, control, motors and drives, photovoltaic systems, and design of magnetic components. The course involves a hands-on laboratory and a substantial final project. Formerly EE 152. Required: EE 101B, EE 102A, EE 108. Recommended: ENGR 40 or EE 122A.

EE258 - INTRO TO RADAR REMOTE SENSING

Credits: 4

Introduction to the principles behind, and applications of, radar as a remote sensing tool. Radar observables and the radar equation, system and subsystem design, signal processing and matched filters, detection problems, radar imaging, range-Doppler processing, interaction of radar waves with Earth or planetary surfaces, interferometers. Applications include polarimetry for surface characterization, measurement of topography and surface change, moving object detection and motion measurements. Graduate/Advanced undergraduate level. Undergraduate students should enroll for 4 units, and graduate students should enroll for 3 units. Prerequisites: deterministic signal processing (EE 102A + B or equivalent); probability and estimation (EE 178 or equivalent).

EE259 - PRINCIPLES SENSING FOR AUTONOMOM

Credits: 3

Basic principles of design and operation of sensors for autonomous systems. Global positioning system (GPS), inertial measurement unit (IMU), Ultrasonic sensor, camera, radar and lidar. Hardware architecture and signal processing algorithms for different sensors. Analysis of sensor performance under different operating conditions, and practical design tradeoffs. Sensor registration and calibration methods. Sensor fusion techniques. Prerequisites: EE 101B or equivalent, EE 102B or equivalent, EE 42 or equivalent, ENGR 108 or equivalent, basic programming (Python, Julia, MATLAB or equivalent).

EE260A - PRINCIPLES OF ROBOT AUTONOMY I

Credits: 4

Basic principles for endowing mobile autonomous robots with perception, planning, and decision-making capabilities. Algorithmic approaches for robot perception, localization, and simultaneous localization and mapping; control of non-linear systems, learning-based control, and robot motion planning; introduction to methodologies for reasoning under uncertainty, e.g., (partially observable) Markov decision processes. Extensive use of the Robot Operating System (ROS) for demonstrations and hands-on activities. Prerequisites: CS 106A or equivalent, CME 100 or equivalent (for linear algebra), and CME 106 or equivalent (for probability theory).

EE261 - FOURIER TRANSFORM/APPLICATIONS

Credits: 3

The Fourier transform as a tool for solving physical problems. Fourier series, the Fourier transform of continuous and discrete signals and its properties. The Dirac delta, distributions, and generalized transforms. Convolutions and correlations and applications; probability distributions, sampling theory, filters, and analysis of linear systems. The discrete Fourier transform and the FFT algorithm. Multidimensional Fourier transform and use in imaging. Further applications to optics, crystallography. Emphasis is on relating the theoretical principles to solving practical engineering and science problems. Prerequisites: Math through ODEs, basic linear algebra, Comfort with sums and discrete signals, Fourier series at the level of 102A

EE263 - INTRO LINEAR DYNAMICAL SYSTEMS

Credits: 3

Applied linear algebra and linear dynamical systems with applications to circuits, signal processing, communications, and control systems. Topics: least-squares approximations of over-determined equations, and least-norm solutions of underdetermined equations. Symmetric matrices, matrix norm, and singular-value decomposition. Eigenvalues, left and right eigenvectors, with dynamical interpretation. Matrix exponential, stability, and asymptotic behavior. Multi-input/multi-output systems, impulse and step matrices; convolution and transfer-matrix descriptions. Control, reachability, and state transfer; observability and least-squares state estimation. Prerequisites: Linear algebra and matrices as in ENGR 108 or MATH 104; ordinary differential equations and Laplace transforms as in EE 102B or CME 102.

EE25N - SCIENCE OF INFORMATION

Credits: 4

We live in the Information Age, but what is information, anyway? In 1948, Claude Shannon published a seminal paper formalizing our modern notion of information. Through lectures and lab visits, we'll learn how information can be measured and represented, why bits are the universal currency for information exchange, and how these ideas led to smartphones, the Internet, and more. We will get a glimpse of information elements in other domains, including neural codes of the brain, cryptographic codes, genetic code, quantum information, and even entertainment. As a final project, students will create podcast episodes on one of the topics explored in the course.

EE260B - PRINCIPLES ROBOT AUTONOMY II

Credits: 4

This course teaches advanced principles for endowing mobile autonomous robots with capabilities to autonomously learn new skills and to physically interact with the environment and with humans. It also provides an overview of different robot system architectures. Concepts that will be covered in the course are: Reinforcement Learning and its relationship to optimal control, contact and dynamics models for prehensile and non-prehensile robot manipulation, imitation learning and human intent inference, as well as different system architectures and their verification. Students will earn the theoretical foundations for these concepts and implement them on mobile manipulation platforms. In homeworks, the Robot Operating System (ROS) will be used extensively for demonstrations and hands-on activities. Prerequisites: CS106A or equivalent, CME 100 or equivalent (for linear algebra), CME 106 or equivalent (for probability theory), and AA 171/274.

EE262 - THREE DIMENSIONAL IMAGING

Credits: 3

Multidimensional time and frequency representations, generalization of Fourier transform methods to non-Cartesian coordinate systems, Hankel and Abel transforms, line integrals, impulses and sampling, reconstruction tomography, imaging radar. The projection-slice and layergram reconstruction methods as developed in radio interferometry. Radar imaging and backprojection algorithms for 3- and 4-D imaging. In weekly labs students create software to form images using these techniques with actual data. Final project consists of design, analysis and simulation of an advanced imaging system. Prerequisites: None required, but recommend EE103, EE261, EE278, some inverse method concepts such as from Geophys281.

EE264 - DIGITAL SIGNAL PROCESSING

Credits: 4

Digital signal processing (DSP) techniques and design of DSP applications. Topics include: discrete-time random signals; sampling and multi-rate systems; oversampling and quantization in A-to-D conversion; properties of LTI systems; quantization in fixed-point implementations of filters; digital filter design; discrete Fourier Transform and FFT; spectrum analysis using the DFT; parametric signal modeling and adaptive filtering. The course also covers applications of DSP in areas such as speech, audio and communication systems. The optional lab section (Section 02) provides a hands-on opportunity to explore the application of DSP theory to practical real-time applications in an embedded processing platform. See ee264.stanford.edu for more information. Register in Section 02 to take the lab. Undergraduate students taking the lab should register for 4 units to meet the EE design requirement. The optional lab section is not available to remote SCPD students. Prerequisites: EE 102A and EE 102B or equivalent, basic programming skills (Matlab and C++)

EE264P - DSP PROJECTS

Credits: 3

This is a companion course to EE 264 Digital Signal Processing for students interested in developing advanced DSP projects beyond the scope of the one credit hour EE 264 lab option (section 2). Weekly meetings with the instructor to plan the week ahead and to share results from the previous are mandatory and will be scheduled at a mutually convenient time. A final project report, project demonstration, and presentation is required. Instructor will determine appropriate number of units based on the project complexity. Prerequisite: EE 264 and instructor approval.

EE267 - VIRTUAL REALITY

Credits: 4

OpenGL, real-time rendering, 3D display systems, display optics & electronics, IMUs and sensors, tracking, haptics, rendering pipeline, multimodal human perception and depth perception, stereo rendering, presence. Emphasis on VR technology. Hands-on programming assignments. The 3-unit version requires a final programming assignment in which you create your own virtual environment. The 4-unit version requires a final course project and written report in lieu of the final assignment. Prerequisites: Strong programming skills, ENGR 108 or equivalent. Helpful: basic computer graphics / OpenGL.

EE268 - ENGR ECON ELECTRICITY MARKETS

Credits: 3

This course presents the power system engineering and economic concepts necessary to understand the costs and benefits of transitioning to a low carbon electricity supply industry. The technical characteristics of generation units and transmission and distribution networks as well as the mechanisms used to operate the electricity supply industries will be studied. The fundamental economics of wholesale markets and how intermittent renewables impact the price and quantity of physical and financial products traded in these markets (e.g., energy, capacity, ancillary services, and financial contracts) will be analyzed. Long-term resource adequacy mechanisms will be introduced and their properties analyzed. The role of both short-duration and seasonal energy storage will be analyzed. Mechanisms for determining the engineering and economic need for transmission network expansions in a wholesale market will be discussed. The impact of distributed versus grid-scale generation on the performance of electricity supply industries will be studied. A detailed treatment of electricity retailing will focus on the importance of active demand-side participation in a low carbon energy sector.

EE26N - WIRELESS WORLD

Credits: 3

The world is increasingly based on wireless communication. Cell phones and WiFi are the most visible examples. Others are key fobs, water meters, gas and electric meters, garage door openers, baby monitors, and the list continues to expand. All of these produce RF signals you can detect and often decode. This seminar will explore how much information you broadcast throughout your day, and how it can easily be received and decoded using inexpensive hardware and public domain software. You will be able to explain why different information services use different frequencies, why they encode the information the way they do, and what security risks they present.

EE264W - DIGITAL SIGNAL PROCESSING

Credits: 5

Writing in the Major (WIM) version of the 4-unit EE 264 theory + lab course. Digital signal processing (DSP) techniques and design of DSP applications. Topics include: discrete-time random signals; sampling and multi-rate systems; oversampling and quantization in A-to-D conversion; properties of LTI systems; quantization in fixed-point implementations of filters; digital filter design; discrete Fourier Transform and FFT; spectrum analysis using the DFT; parametric signal modeling and adaptive filtering. The course also covers applications of DSP in areas such as speech, audio and communication systems. The lab component provides a hands-on opportunity to explore the application of DSP theory to practical real-time applications in an embedded processing platform. See ee264.stanford.edu for more information. Prerequisites: EE 102A and EE 102B or equivalent, basic programming skills (Matlab and C++)

EE267W - VIRTUAL REALITY (WIM)

Credits: 5

Writing in the Major (WIM) version of the 4-unit EE 267 theory + lab/project course. This course also meets the EE design requirement. Topics include: OpenGL, real-time rendering, 3D display systems, display optics & electronics, IMUs and sensors, tracking, haptics, rendering pipeline, multimodal human perception and depth perception, stereo rendering, presence. Emphasis on VR technology. Hands-on programming assignments. The 5-unit WIM version requires everything the 4-unit version does, i.e. a final course project and written report in lieu of the final assignment. The 5-unit WIM version additionally requires participation in 2 writing in the major workshops, and weekly writing assignments. Prerequisites: Strong programming skills, ENGR 108 or equivalent. Helpful: basic computer graphics / OpenGL.

EE269 - SIGNAL PROCESSING FOR ML

Credits: 3

This course will introduce you to fundamental signal processing concepts and tools needed to apply machine learning to discrete signals. You will learn about commonly used techniques for capturing, processing, manipulating, learning and classifying signals. The topics include: mathematical models for discrete-time signals, vector spaces, Fourier analysis, time-frequency analysis, Z-transforms and filters, signal classification and prediction, basic image processing, compressed sensing and deep learning. This class will culminate in a final project. Prerequisites: EE 102A and EE 102B or equivalent, basic programming skills (Matlab). ENGR 108 and EE 178 are recommended.

EE270 - LARGE SCALE MATRIX COMPUTATION

Credits: 3

Massive data sets are now common to many different fields of research and practice. Classical numerical linear algebra can be prohibitively costly in many modern problems. This course will explore the theory and practice of randomized matrix computation and optimization for large-scale problems to address challenges in modern massive data sets. Applications in machine learning, statistics, signal processing and data mining will be surveyed. Prerequisites: familiarity with linear algebra (ENGR 108 or equivalent), basic probability and statistics (EE 178 or equivalent), basic programming skills.

EE271 - INTRODUCTION TO VLSI SYSTEMS

Credits: 3

Provides a quick introduction to MOS transistors and IC fabrication and then creates abstractions to allow you to create and reason about complex digital systems. It uses a switch resistor model of a transistor, uses it to model gates, and then shows how gates and physical layout can be synthesized from Verilog or SystemVerilog descriptions. Most of the class will be spent on providing techniques to create designs that can be validated, are low power, provide good performance, and can be completed in finite time. Prerequisites: 101A and 108; familiarity with transistors, logic design, Verilog and digital system organization

EE273 - DIGITAL SYSTEMS ENGINEERING

Credits: 3

Electrical issues in the design of high-performance digital systems, including signaling, timing, synchronization, noise, and power distribution. High-speed signaling methods; noise in digital systems, its effect on signaling, and methods for noise reduction; timing conventions; timing noise (skew and jitter), its effect on systems, and methods for mitigating timing noise; synchronization issues and synchronizer design; clock and power distribution problems and techniques; impact of electrical issues on system architecture and design. Prerequisites: EE101A and EE108A. Recommended: EE114/214A.

EE276 - INFORMATION THEORY

Credits: 3

(Formerly EE 376A.) Project-based course about how to measure, represent, and communicate information effectively. Why bits have become the universal currency for information exchange. How information theory bears on the design and operation of modern-day systems such as smartphones and the Internet. The role of entropy and mutual information in data compression, communication, and inference. Practical compressors and error correcting codes. The information theoretic way of thinking. Relations and applications to probability, statistics, machine learning, biological and artificial neural networks, genomics, quantum information, and blockchains. Prerequisite: a first undergraduate course in probability.

EE278 - STATISTICAL SIGNAL PROCESSING

Credits: 3

Review of basic probability and random variables. Random vectors and processes; convergence and limit theorems; IID, independent increment, Markov, and Gaussian random processes; stationary random processes; autocorrelation and power spectral density; mean square error estimation, detection, and linear estimation. Formerly EE 278B. Prerequisites: EE178 and linear systems and Fourier transforms at the level of EE102A,B or EE261.

EE272 - DESIGN PROJECTS IN VLSI SYSTEM

Credits: 4

This course will introduce you to mixed signal design and the electronic design automation (EDA) tools used for it. Working in teams, you will create a chip with a digital deep neural network (DNN) accelerator and a small analog block using a modern design flow and EDA tools. The project involves writing a synthesizable C++ and a Verilog model of your chip, creating a testing/debug strategy for your chip, wrapping custom layout to fit into a standard cell system, using synthesis and place and route tools to create the layout of your chip, and understanding all the weird stuff you need to do to tape-out a chip. Useful for anyone who will build a chip in their Ph.D. Pre-requisites: EE271 and experience in digital/analog circuit design.

EE274 - DATA COMPRESSION

Credits: 3

The course focuses on the theory and algorithms underlying modern data compression. The first part of the course introduces techniques for entropy coding and for lossless compression. The second part covers lossy compression including techniques for multimedia compression. The last part of the course will cover advanced theoretical topics and applications, such as neural network based compression, distributed compression, and computation over compressed data. Prerequisites: basic probability and programming background (EE178, CS106B or equivalent), a course in signals and systems (EE102A), or instructor's permission.

EE277 - REINFORCEMENT LEARNING

Credits: 3

Reinforcement learning addresses the design of agents that improve decisions while operating within complex and uncertain environments. This course covers principled and scalable approaches to realizing a range of intelligent learning behaviors. Topics include environment models, planning, abstraction, prediction, credit assignment, exploration, and generalization. Motivating examples will be drawn from web services, control, finance, and communications. Prerequisites: programming (e.g., CS106B), probability (e.g., MS&E 121, EE 178 or CS 109), machine learning (e.g., EE 104/CME 107, MS&E 226 or CS 229).

EE279 - INTRO TO DIGITAL COMMUNICATION

Credits: 3

Digital communication is a rather unique field in engineering in which theoretical ideas have had an extraordinary impact on the design of actual systems. The course provides a basic understanding of the analysis and design of digital communication systems, building on various ideas from probability theory, stochastic processes, linear algebra and Fourier analysis. Topics include: detection and probability of error for binary and M-ary signals (PAM, QAM, PSK), receiver design and sufficient statistics, controlling the spectrum and the Nyquist criterion, bandpass communication and up/down conversion, design trade-offs: rate, bandwidth, power and error probability, coding and decoding (block codes, convolutional coding and Viterbi decoding). Prerequisites: 179 or 261, and 178 or 278

EE282 - COMPUTER SYSTEMS ARCHITECTURE

Credits: 3

Course focuses on how to build modern computing systems, namely notebooks, smartphones, and data centers, covering primarily their hardware architecture and certain system software aspects. For each system class, we cover the system architecture, processor technology, advanced memory hierarchy and I/O organization, power and energy management, and reliability. We will also cover topics such as interactions with system software, virtualization, solid state storage, and security. The programming assignments allow students to explore performance/energy tradeoffs when using heterogeneous hardware resources on smartphone devices. Prerequisite: EE180. Recommended: CS 140.

EE284A - INTERNET OF THINGS

Credits: 3

Internet of Things (IoT) origin, vision and definition. Application domains, use case scenarios and value propositions. Functional blocks of IoT systems: devices, communications, services, management, security, and application. Architectural reference model and design methodology. IoT Devices: sensors, actuators and embedded systems. Communications aspects of IoT systems: Internet infrastructure; wireless local area networks; radio access networks; wireless personal area networks; wireless sensor networks; wireless communication in vehicular environments; 5G. Current IoT frameworks and underlying architectures. Data storage and analytics. Web services. IoT system management tools. Security aspects of IoT systems. Open issues.

EE290A - CURR PRACT TRAINING ELEC ENGR

Credits: 1

For EE majors who need work experience as part of their program of study. Final report required. Prerequisites: for 290B, EE MS and PhD students who have received a Satisfactory ("S") grade in EE290A; for 290C, EE PhD degree candidacy and an "S" grade in EE 290B; for 290D, EE PhD degree candidacy, an "S" grade in EE 290C and instructor consent.

EE290C - CURR PRACT TRAINING ELEC ENGR

Credits: 1

For EE majors who need work experience as part of their program of study. Final report required. Prerequisites: for 290B, EE MS and PhD students who have received a Satisfactory ("S") grade in EE290A; for 290C, EE PhD degree candidacy and an "S" grade in EE 290B; for 290D, EE PhD degree candidacy, an "S" grade in EE 290C and instructor consent.

EE290E - CURR PRACT TRAINING ELEC ENGR

Credits: 1

For EE majors who need work experience as part of their program of study. Final report required. Prerequisites: for 290B, EE MS and PhD students who have received a Satisfactory ("S") grade in EE290A; for 290C, EE PhD degree candidacy and an "S" grade in EE 290B; for 290D, EE PhD degree candidacy, an "S" grade in EE 290C and instructor consent; for 290E, EE PhD degree candidacy, an "S" grade in EE290D and instructor consent.

EE290G - CURR PRACT TRAINING ELEC ENGR

Credits: 1

For EE majors who need work experience as part of their program of study. Final report required. Prerequisites: EE PhD degree candidacy, an "S" grade in EE290F and instructor consent.

EE284 - INTRODUCTION COMPUTER NETWORKS

Credits: 3

Structure and components of computer networks; functions and services; packet switching; layered architectures; OSI reference model; physical layer; data link layer; error control; window flow control; media access control protocols used in local area networks (Ethernet, Token Ring, FDDI) and satellite networks; network layer (datagram service, virtual circuit service, routing, congestion control, Internet Protocol); transport layer (UDP, TCP); application layer.

EE285 - EMBEDDED SYSTEMS WORKSHOP

Credits: 9

Project-centric building hardware and software for embedded computing systems. This year the course projects are on a large interactive light sculpture to be installed in Packard. Syllabus topics will be determined by the needs of the enrolled students and projects. Examples of topics include: interrupts and concurrent programming, mechanical control, state-based programming models, signaling and frequency response, mechanical design, power budgets, software, firmware, and PCB design. Interested students can help lead community workshops to begin building the installation. Prerequisites: one of CS107, EE101A, EE108, ME80.

EE290B - CURR PRACT TRAINING ELEC ENGR

Credits: 1

For EE majors who need work experience as part of their program of study. Final report required. Prerequisites: for 290B, EE MS and PhD students who have received a Satisfactory ("S") grade in EE290A; for 290C, EE PhD degree candidacy and an "S" grade in EE 290B; for 290D, EE PhD degree candidacy, an "S" grade in EE 290C and instructor consent.

EE290D - CURR PRACT TRAINING ELEC ENGR

Credits: 1

For EE majors who need work experience as part of their program of study. Final report required. Prerequisites: for 290B, EE MS and PhD students who have received a Satisfactory ("S") grade in EE290A; for 290C, EE PhD degree candidacy and an "S" grade in EE 290B; for 290D, EE PhD degree candidacy, an "S" grade in EE 290C and instructor consent.

EE290F - CURR PRACT TRAINING ELEC ENGR

Credits: 1

For EE majors who need work experience as part of their program of study. Final report required. Prerequisites: EE PhD degree candidacy, an "S" grade in EE 290E and instructor consent.

EE292A - ELECTRONIC DESIGN AUTOMATION

Credits: 3

The class teaches cutting-edge optimization and analysis algorithms for the design of complex digital integrated circuits and their use in designing machine learning hardware. It provides working knowledge of the key technologies in Electronic Design Automation (EDA), focusing on synthesis, placement and routing algorithms that perform the major transformations between levels of abstraction and get a design ready to be fabricated. As an example, the design of a convolutional neural network (CNN) for basic image recognition illustrates the interaction between hardware and software for machine learning. It will be implemented on a state-of-the-art FPGA board. Prerequisite: EE 108.

EE292C - CHEMICAL VAPOR DEPOSITION

Credits: 1

Fundamental aspects of CVD are initially considered, first focusing on processes occurring in the gas phase and then on those occurring on the surface. Qualitative understanding is emphasized, with minimal use of equations. Adding energy both thermally and by using a plasma is discussed; atomic-layer deposition is briefly considered. Examples of CVD equipment are examined. The second portion of the tutorial examines layers deposited by CVD. The focus is on group IV semiconductors especially epitaxial and heteroepitaxial deposition, in which the crystal structure of the depositing layer is related to that of the substrate. Polycrystalline silicon and the IC interconnect system are then discussed. Finally, the use of high-density plasmas for rapid gap filling is contrasted with alternative CVD dielectric deposition processes.

EE292E - IMAGE SYSTEMS ENGINEERING

Credits: 99

Seminar. For engineering students interested in camera and display engineering, computer vision, and computational imaging. Speakers include Stanford faculty and research scientists as well as industry professionals, mostly from consumer electronics companies.

EE292H - ENGRING, ENTR, & CLIMATE CHANG

Credits: 999

The purpose of this seminar series course is to help students and professionals develop the tools to apply the engineering and entrepreneurial mindset to problems that stem from climate change, in order to consider and evaluate possible stabilizing, remedial and adaptive approaches. This course is not a crash course on climate change or policy. Instead we will focus on learning about and discussing the climate problems that seem most tractable to these approaches. Each week Dr. Field and/or a guest speaker will lead a short warm-up discussion/activity and then deliver a talk in his/her area of expertise. We will wrap up with small-group and full-class discussions of related challenges/opportunities and possible engineering-oriented solutions. Class members are asked to do background reading before each class, to submit a question before each lecture, and to do in-class brainstorming. May be repeated for credit.

EE292N - WIRELESS FRONTIERS

Credits: 1

This course aims to raise the interest of senior undergraduate students and junior graduate students to the area of wireless from communication, gesture detection, power delivery to radar applications. It serves as an introduction to wireless through a series of seminars with invited speakers from both industry and academia.

EE292D - EMBEDDED SYSTEMS

Credits: 3

This is a project-based class where students will learn how to develop machine learning models for execution in resource constrained environments such as embedded systems. In this class students will learn about techniques to optimize machine learning models and deploy them on a device such as a Arduino, Raspberry PI, Jetson, or Edge TPUs. The class has a significant project component. Prerequisites: CS 107(required), CS 229 (recommended), CS 230 (recommended).

EE292F - IMAGE PROCESSING OF FINE ART

Credits: 3

This course presents the application of rigorous digital image processing to problems in visualization and understanding of fine paintings, drawings, and other two-dimensional artworks. It builds upon a wealth of techniques but modifies and applies them to cases of interest to the technical art community. Such techniques include transforms such as DCT and wavelets, color quantization, blind source (image) separation, edge detection, super-resolution, visual style learning and transfer, digital in-painting, color transforms, level-set analysis, estimation of region statistics, Affine image transforms, and many others. Students will perform several projects which will involve coding, mathematical/statistical analysis, and explaining the relevance of the work to art scholarship.

EE292I - INSANELY GREAT PRODUCTS

Credits: 1

Great products emerge from a sometimes conflict-laden process of collaboration between different functions within companies. This Seminar seeks to demystify this process via case-studies of successful products and companies. Engineering management and businesspeople will share their experiences in discussion with students. Previous companies profiled: Apple, Intel, Facebook, and Genentech -- to name a few. Previous guests include: Jon Rubinstein (NeXT, Apple, Palm), Diane Greene (VMware), and Ted Hoff (Intel). Pre-requisites: None

EE292Q - 3D+ IMAGING SENSORS

Credits: 3

Introduction to operation principles and key performance aspects of 3D+ imaging sensors used widely in industry. Concepts include imaging physics, data acquisition and image formation methods, and signal and image quality metrics that are broadly applicable across sensor types. Practical examples and demonstrations of various sensors such as radar, acoustic, LIDAR, and ToF modules will be presented in class as well as through structured labs. Invited speakers will highlight emerging 3D+ imaging applications that these sensors are enabling today. Prerequisites: EE 101A or equivalent. EE 102A or equivalent.

EE292T - SMART GRIDS AND ADVANCED POWER

Credits: 4

A series of seminar and lectures focused on power engineering. Renowned researchers from universities and national labs will deliver bi-weekly seminars on the state of the art of power system engineering. Seminar topics may include: power system analysis and simulation, control and stability, new market mechanisms, computation challenges and solutions, detection and estimation, and the role of communications in the grid. The instructors will cover relevant background materials in the in-between weeks. The seminars are planned to continue throughout the next academic year, so the course may be repeated for credit.

EE292Y - EMERGING HARDWARE PLATFORMS

Credits: 3

Research seminar on software techniques for emerging computational substrates with guest lectures from hardware designers from research and industry. This seminar explores the benefits of novel hardware technologies, the challenges gating broad adoption of these technologies, and how software techniques can help mitigate these challenges and improve the usability of these hardware platforms. Note that the computational substrates discussed vary depending on the semester. Topics covered include: In-memory computing platforms, dynamical system-solving mixed-signal devices, exible and bendable electronics, neuromorphic computers, intermittent computing platforms, ReRAMs, DNA-based storage, and optical computing platforms. Prerequisites: CS107 or CS107E (required) and EE180 (recommended).

EE300 - MASTER'S THESIS

Credits: 999

Independent work under the direction of a department faculty. Written thesis required for final letter grade. The continuing grade 'N' is given in quarters prior to thesis submission. See 390 if a letter grade is not appropriate. Course may be repeated for credit.

EE303 - AUTONOMOUS IMPLANTABLE SYSTEMS

Credits: 3

Integrating electronics with sensing, stimulation, and locomotion capabilities into the body will allow us to restore or enhance physiological functions. In order to be able to insert these electronics into the body, energy source is a major obstacle. This course focuses on the analysis and design of wirelessly powered catheter-deliverable electronics. Emphases will be on the interaction between human and electromagnetic fields in order to transfer power to the embedded electronics via electromagnetic fields, power harvesting circuitry, electrical-tissue interface, and sensing and actuating frontend designs.

EE292X - BATTERY SYSTEMS

Credits: 3

Driven by high-capacity battery systems, electrification is transforming mobility solutions and the grid that powers them. This course provides an introduction to battery systems for transportation and grid services: cell technologies, topology selection, thermal and aging management, safety monitoring, AC and DC charging, and operation control/optimization. Invited experts introduce students to the state of the heart of each topic. The course is aimed at mezzanine and graduate levels students who wish to design battery systems, model them from data, integrate them into applications, or just learn about them. It can be taken for 1 unit (Credit/no Credit) for attending seminars, or for 3 units (letter grade only) for also doing an optional project. Prerequisites: No prerequisites needed for taking the course for 1 unit. Relevant background in selected project area is recommended, for example, CEE 272R for grid applications; EE 253 for AC or DC charging and battery controller design; CEE 322, CS 229 or EE 104 for data-based projects.

EE293B - FUNDAMENTALS OF ENERGY PROCESS

Credits: 3

For seniors and graduate students. Covers scientific and engineering fundamentals of renewable energy processes involving heat. Thermodynamics, heat engines, solar thermal, geothermal, biomass. Recommended: MATH 19-21; PHYSICS 41, 43, 45

EE301 - INTRODUCTORY RESEARCH SEMINAR

Credits: 4

The EE 301 seminar course is offered primarily for incoming EE PhD students; however, all graduate or undergraduate students may enroll, and all students and faculty in the Department are welcome to attend. EE faculty members present seminars on their research, giving new PhD students an overview of research opportunities across the Department.

EE308 - ADVANCED CIRCUIT TECHNIQUES

Credits: 3

Design of advanced analog circuits at the system level, including switching power converters, amplitude-stabilized and frequency-stabilized oscillators, voltage references and regulators, power amplifiers and buffers, sample-and-hold circuits, and application-specific op-amp compensation. Approaches for finding creative design solutions to problems with difficult specifications and hard requirements. Emphasis on feedback circuit techniques, design-oriented thinking, and hands-on experience with modern analog building blocks. Several designs will be built and evaluated, along with associated laboratory projects. Prerequisite: EE 251 or EE 314A.

EE309A - SEMICOND MEM DEVICES & CIRCUIT

Credits: 3

The functionality and performance of ULSI systems are increasingly dependent upon the characteristics of the memory subsystem. This course introduces students to various semiconductor memory devices: SRAM, DRAM and FLASH, that are used in today's memory subsystems. The course will cover various aspects of semiconductor memories, including basic operation principles, device design considerations, device scaling, device fabrication, memory array architecture, and addressing and readout circuits. The course will also introduce students to recent research in near- and in-memory computing using these memory technologies. The next course in this series is EE 309B, which talks about emerging non-volatile memory devices and circuit design. Pre-requisite: EE 216. Preferred: EE 316.

EE310 - SYSTEMX: UBIQUITOUS SENSING, C

Credits: 999

This is a seminar course with invited speakers. Sponsored by Stanford's SystemX Alliance, the talks will cover emerging topics in contemporary hardware/software systems design. Special focus will be given to the key building blocks of sensors, processing elements and wired/wireless communications, as well as their foundations in semiconductor technology, SoC construction, and physical assembly as informed by the SystemX Focus Areas. The seminar will draw upon distinguished engineering speakers from both industry and academia who are involved at all levels of the technology stack and the applications that are now becoming possible. May be repeat for credit

EE312 - INTEGRATED CIRCUIT FABRIC LAB

Credits: 4

Formerly EE 410. Fabrication, simulation, and testing of a submicron CMOS process. Practical aspects of IC fabrication including silicon wafer cleaning, photolithography, etching, oxidation, diffusion, ion implantation, chemical vapor deposition, physical sputtering, and electrical testing. Students also simulate the CMOS process using process simulator TSUPREM4 of the structures and electrical parameters that should result from the process flow. Taught in the Stanford Nanofabrication Facility (SNF). Preference to students pursuing doctoral research program requiring SNF facilities. Enrollment limited to 20. Prerequisites: EE 212, EE 216, or consent of instructor.

EE315 - ANALOG-DIGITAL INTERFACE CIRCU

Credits: 3

Analysis and design of circuits and circuit architectures for signal conditioning and data conversion. Fundamental circuit elements such as operational transconductance amplifiers, active filters, sampling circuits, switched capacitor stages and voltage comparators. Sensor interfaces for micro-electromechanical and biomedical applications. Nyquist and oversampling A/D and D/A converters. Prerequisite: EE 214B.

EE309B - NON-VOLATILE MEM DEVICE

Credits: 3

The functionality and performance of ULSI systems are increasingly dependent upon the characteristics of the memory subsystem. This course starts off where EE 309A leaves, and introduces students to various emerging non-volatile memory devices: metal oxide resistive switching memory (RRAM), nanoconductive bridge memory (CBRAM), phase change memory (PCM), magnetic tunnel junction memory, spin-transfer-torque random access memory (MRAM, STT-RAM), ferroelectric memory (FRAM) and ferroelectric transistor (FeFET). For each of these memories, the course will cover basic operation principles, device design considerations, device scaling, device fabrication, memory array architecture, and addressing and readout circuits. The course will also introduce students to recent in-memory computing research using these memory technologies. Pre-requisite: EE 216. Preferred: EE 316, EE 309A.

EE311 - ADV INTEGR CIRCUIT TECHNOLOGY

Credits: 3

What are the practical and fundamental limits to the evolution of the technology of modern MOS devices and interconnects? How are modern devices and circuits fabricated and what future changes are likely? Advanced techniques and models of MOS devices and back-end (interconnect and contact) processing. What are future device structures and materials to maintain progress in integrated electronics? MOS front-end and back-end process integration. Prerequisites: EE 216 or equivalent. Recommended: EE 212.

EE314A - RF CIRCUIT DESIGN

Credits: 3

Design of RF integrated circuits for communications systems, primarily in CMOS. Topics: the design of matching networks and low-noise amplifiers at RF, mixers, modulators, and demodulators; review of classical control concepts necessary for oscillator design including PLLs and PLL-based frequency synthesizers. Design of low phase noise oscillators. Design of high-efficiency (e.g., class E, F) RF power amplifiers, coupling networks. Behavior and modeling of passive and active components at RF. Narrowband and broadband amplifiers; noise and distortion measures and mitigation methods. Overview of transceiver architectures. Prerequisite: EE 214B.

EE316 - ADVANCED VLSI DEVICES

Credits: 3

In modern VLSI technologies, device electrical characteristics are sensitive to structural details and therefore to fabrication techniques. How are advanced VLSI devices designed and what future changes are likely? What are the implications for device electrical performance caused by fabrication techniques? Physical models for nanometer scale structures, control of electrical characteristics (threshold voltage, short channel effects, ballistic transport) in small structures, and alternative device structures for VLSI. Prerequisites: 216 or equivalent. Recommended: EE 212.

EE317 - WIDE-BANDGAP DEVICE

Credits: 3

Wide-bandgap (WBG) semiconductors present a pathway to push the limits of efficiency in optoelectronics and electronics enabling significant energy savings, offering new and compact architecture, and more functionality. We will first study the examples set by GaN and SiC in lighting, radiofrequency and power applications, then use it to explore new materials like Ga₂O₃, AlN and diamond to understand their potential to drive the future semiconductor industry. The term papers will include a short project that may require simulation to conduct device design and analysis. Prerequisites: EE216 or EE218

EE323 - ENERGY IN ELECTRONICS

Credits: 3

EE 323 examines energy in modern nanoelectronics, from fundamentals to systems. Fundamental topics include energy storage and transfer via electrons and phonons, ballistic limits of current and heat, meso- to macroscale mobility and thermal conductivity. Applied topics include power in nanoscale devices (1D nanotubes and nanowires, 2D materials, 3D silicon CMOS, resistive memory and interconnects), circuit leakage, temperature measurements, thermoelectric energy conversion, and thermal challenges in densely integrated systems. Basic knowledge of semiconductors, transistors, and Matlab (or similar) are recommended.

EE329 - ELECTRONIC STRUCTURE SURFACE

Credits: 3

Physical concepts and phenomena for surface science techniques probing the electronic and chemical structure of surfaces, interfaces and nanomaterials. Microscopic and atomic models of microstructures; applications including semiconductor device technology, catalysis and energy. Physical processes of UV and X-ray photoemission spectroscopy, Auger electron spectroscopy, surface EXAFS, low energy electron diffraction, electron/photon stimulated ion desorption, scanning tunneling spectroscopy, ion scattering, energy loss spectroscopy and related imaging methods; and experimental aspects of these surface science techniques. Prerequisites: PHYSICS 70 and MATSCI 199/209, or consent of instructor.

EE336 - NANOPHOTONICS

Credits: 3

Recent developments in micro- and nanophotonic materials and devices. Basic concepts of photonic crystals. Integrated photonic circuits. Photonic crystal fibers. Superprism effects. Optical properties of metallic nanostructures. Sub-wavelength phenomena and plasmonic excitations. Meta-materials. Prerequisite: Electromagnetic theory at the level of 242.

EE346 - INTRO TO NONLINEAR OPTICS

Credits: 3

Wave propagation in anisotropic, nonlinear, and time-varying media. Microscopic and macroscopic description of electric-dipole susceptibilities. Free and forced waves; phase matching; slowly varying envelope approximation; dispersion, diffraction, space-time analogy. Harmonic generation; frequency conversion; parametric amplification and oscillation; electro-optic light modulation. Raman and Brillouin scattering; nonlinear processes in optical fibers. Prerequisites: 242, 236C.

EE320 - NANO-ELECTRONICS

Credits: 3

This course covers the device physics and operation principles of nanoelectric devices, with a focus on devices for energy-efficient computation. Topics covered include devices based on new nanomaterials such as carbon nanotubes, semiconductor nanowires, and 2D layered materials such as graphene; non-FET based devices such as nanoelectromechanical (NEM) relay, single electron transistors (SET) and resonant tunneling diodes (RTD); as well as FET-based devices such as tunnel FET. Devices targeted for both logic and memory applications are covered. Prerequisites: Undergraduate device physics, EE222, EE216, EE316. Recommended courses: EE223, EE228, EE311.

EE327 - PROP SEMICONDUCTOR MATERIALS

Credits: 3

Modern semiconductor devices and integrated circuits are based on unique energy band, carrier transport, and optical properties of semiconductor materials. How to choose these properties for operation of semiconductor devices. Emphasis is on quantum mechanical foundations of the properties of solids, energy bandgap engineering, semi-classical transport theory, semi-conductor statistics, carrier scattering, electro-magneto transport effects, high field ballistic transport, Boltzmann transport equation, quantum mechanical transitions, optical absorption, and radiative and non-radiative recombination that are the foundations of modern transistors and optoelectronic devices. Prerequisites: EE216 or equivalent.

EE332 - LASER DYNAMICS

Credits: 3

Dynamic and transient effects in lasers including spiking, Q-switching, mode locking, frequency modulation, frequency and spatial mode competition, linear and nonlinear pulse propagation, pulse shaping. Formerly EE232. Prerequisite: 236C.

EE340 - OPTICAL MICRO AND NANO

Credits: 3

Optical micro- and nano-cavities and their device applications. Types of optical cavities (microdisks, microspheres, photonic crystal cavities, plasmonic cavities), and their electromagnetic properties, design, and fabrication techniques. Cavity quantum electrodynamics: strong and weak-coupling regime, Purcell factor, spontaneous emission control. Applications of optical cavities, including low-threshold lasers, optical modulators, quantum information processing devices, and bio-chemical sensors. Prerequisites: Advanced undergraduate or basic graduate level knowledge of electromagnetics, quantum.

EE347 - OPTICAL METHODS ENGR SCIENCE

Credits: 3

Design and understanding of modern optical systems. Topics: geometrical optics; aberration theory; systems layout; applications such as microscopes, telescopes, optical processors. Computer ray tracing program as a design tool. Prerequisite: 236A or equivalent.

EE348 - ADV OPTICAL FIBER COMMUNICATIO

Credits: 3

Optical amplifiers: gain, saturation, noise. Semiconductor amplifiers. Erbium-doped fiber amplifiers. System applications: preamplified receiver performance, amplifier chains. Raman amplifiers, lumped vs. distributed amplification. Group-velocity dispersion management: dispersion-compensating fibers, filters, gratings. Interaction of dispersion and nonlinearity, dispersion maps. Multichannel systems. Wavelength-division multiplexing components: filters, multiplexers. WDM systems, crosstalk. Time, subcarrier, code and polarization-division multiplexing. Comparison of modulation techniques: differential phase-shift keying, phase-shift keying, quadrature-amplitude modulation. Comparison of detection techniques: noncoherent, differentially coherent, coherent. Prerequisite: 247.

EE355 - IMAGING RADAR + APPLICATIONS

Credits: 3

Radar remote sensing, radar image characteristics, viewing geometry, range coding, synthetic aperture processing, correlation, range migration, range/Doppler algorithms, wave domain algorithms, polar algorithm, polarimetric processing, interferometric measurements. Applications: surface deformation, polarimetry and target discrimination, topographic mapping surface displacements, velocities of ice fields. Prerequisites: EE261. Recommended: EE254, EE278, EE279.

EE356B - MAGNETICS DESIGN IN POWER ELEC

Credits: 3

Inductors and transformers are ubiquitous components in any power electronics system. They are components that offer great design flexibility, provide electrical isolation and can reduce semiconductor stresses, but they often dominate the size and cost of a power converter and are notoriously difficult to miniaturize. In this class we will discuss the design and modeling of magnetic components, which are essential tasks in the development of high performance converters and study advanced applications. Prerequisites: EE153/EE253.

EE359 - WIRELESS COMMUNICATIONS

Credits: 4

This course will cover advanced topics in wireless communications as well as current wireless system design. Topics include: an overview of current and future wireless systems; wireless channel models including path loss, shadowing, and statistical multipath channel models; fundamental capacity limits of wireless channels; digital modulation and its performance in fading and under intersymbol interference; techniques to combat fading including adaptive modulation and diversity; multiple antenna (MIMO) techniques to increase capacity and diversity, intersymbol interference including equalization, multicarrier modulation (OFDM), and spread spectrum; and multiuser system design, including multiple access techniques. Course is 3 units but can be taken for 4 units with an optional term project. Prerequisite: 279 or instructor consent.

EE349 - ADVANCED TOPICS IN NANO-OPTICS

Credits: 3

Electromagnetic phenomena at the nanoscale. Dipolar interactions between emitters and nanostructures, weak and strong coupling, surface plasmon polaritons and localized plasmons, electromagnetic field enhancements, and near-field coupling between metallic nanostructures. Numerical tools will be taught and used to simulate nano-optical phenomena. Prerequisite: EE242 or equivalent.

EE356A - RESONANT CONVERTERS

Credits: 3

Miniaturization of efficient power converters remain a challenge in power electronics whose goal is improving energy use and reducing waste. In this course, we will study the design of Resonant converters which are capable of operating at higher frequencies than their 'hard-switch' counterparts. Resonant converter are found in high performance applications where high control bandwidth and high power density are required. We will also explore practical design issues and trade off in selecting converter topologies in high performance applications. Prerequisites: EE153/EE253.

EE358 - WIRELESS SYSTEM DESIGN

Credits: 3

Wireless systems are commonly used in our day-to-day life. Different applications impose different design trade-offs and optimizations. This course will cover various building blocks (filters, channel coding, MIMO algorithms, carrier/timing recovery, and preamble design) of a complete wireless system and their respective design trade-offs. Students will implement these building blocks in Simulink and software defined radio to enhance their understandings. The course will also cover various wireless standards, RF chain and analog-digital co-design, digital implementation platforms, and DSP arithmetic. Prerequisites: One of EE 279, EE 359, EE 379, or equivalent.

EE364A - CONVEX OPTIMIZATION I

Credits: 3

Convex sets, functions, and optimization problems. The basics of convex analysis and theory of convex programming: optimality conditions, duality theory, theorems of alternative, and applications. Least-squares, linear and quadratic programs, semidefinite programming, and geometric programming. Numerical algorithms for smooth and equality constrained problems; interior-point methods for inequality constrained problems. Applications to signal processing, communications, control, analog and digital circuit design, computational geometry, statistics, machine learning, and mechanical engineering. Prerequisite: linear algebra such as EE263, basic probability.

EE364B - CONVEX OPTIMIZATION II

Credits: 3

Continuation of 364A. Subgradient, cutting-plane, and ellipsoid methods. Decentralized convex optimization via primal and dual decomposition. Monotone operators and proximal methods; alternating direction method of multipliers. Exploiting problem structure in implementation. Convex relaxations of hard problems. Global optimization via branch and bound. Robust and stochastic optimization. Applications in areas such as control, circuit design, signal processing, and communications. Course requirements include project. Prerequisite: 364A.

EE368 - DIGITAL IMAGE PROCESSING

Credits: 3

Image sampling and quantization color, point operations, segmentation, morphological image processing, linear image filtering and correlation, image transforms, eigenimages, multiresolution image processing, noise reduction and restoration, feature extraction and recognition tasks, image registration. Emphasis is on the general principles of image processing. Students learn to apply material by implementing and investigating image processing algorithms in Matlab and optionally on Android mobile devices. Term project. Recommended: EE261, EE278.

EE369B - MEDICAL IMAGING SYSTEMS II

Credits: 3

Imaging internal structures within the body using magnetic resonance studied from a systems viewpoint. Analysis of magnetic resonance imaging systems including physics, Fourier properties of image formation, effects of system imperfections, image contrast, and noise. Pre- or corequisite: EE261 or equivalent

EE371 - ADVANCED VLSI CIRCUIT DESIGN

Credits: 3

Design of high-performance digital systems, the things that cause them to fail, and how to avoid these problems. Topics will focus on current issues including: wiring resistance and how to deal with it, power and Gnd noise and regulation, clock (or asynchronous) system design and how to minimize clocking overhead, high-speed I/O design, energy minimization including leakage control, and structuring your Verilog code to result in high-performance, low energy systems. Extensive use of modern CAD tools. Prerequisites: EE 213 and EE 271, or consent of instructor.

EE373A - ADAPTIVE SIGNAL PROCESSING

Credits: 3

Learning algorithms for adaptive digital filters. Self-optimization. Wiener filter theory. Quadratic performance functions, their eigenvectors and eigenvalues. Speed of convergence. Asymptotic performance versus convergence rate. Applications of adaptive filters to statistical prediction, process modeling, adaptive noise canceling, adaptive antenna arrays, adaptive inverse control, and equalization and echo canceling in modems. Artificial neural networks. Cognitive memory/human and machine. Natural and artificial synapses. Hebbian learning. The Hebbian-LMS algorithm. Theoretical and experimental research projects in adaptive filter theory, communications, audio systems, and neural networks. Biomedical research projects, supervised jointly by EE and Medical School faculty. Recommended: EE263, EE264, EE278.

EE367 - COMPUTATIONAL IMAGING

Credits: 3

Digital photography and basic image processing, convolutional neural networks for image processing, denoising, deconvolution, single pixel imaging, inverse problems in imaging, proximal gradient methods, introduction to wave optics, time-of-flight imaging, end-to-end optimization of optics and imaging processing. Emphasis is on applied image processing and solving inverse problems using classic algorithms, formal optimization, and modern artificial intelligence techniques. Students learn to apply material by implementing and investigating image processing algorithms in Python. Term project. Recommended: EE261, EE263, EE278.

EE369A - MEDICAL IMAGING SYSTEMS I

Credits: 3

Imaging internal structures within the body using high-energy radiation studied from a systems viewpoint. Modalities covered: x-ray, computed tomography, and nuclear medicine. Analysis of existing and proposed systems in terms of resolution, frequency response, detection sensitivity, noise, and potential for improved diagnosis. Prerequisite: EE 261

EE369C - MEDICAL IMAGE RECONSTRUCTION

Credits: 3

Reconstruction problems from medical imaging, including magnetic resonance imaging (MRI), computed tomography (CT), and positron emission tomography (PET). Problems include reconstruction from non-uniform frequency domain data, automatic deblurring, phase unwrapping, reconstruction from incomplete data, and reconstruction from projections. Prerequisite: 369B.

EE372 - DESIGN PROJECTS IN VLSI SYSTEM

Credits: 5

This is a follow on course to EE272. While in EE272 you learn the EDA tool flow and design a pre-specified digital neural network accelerator and an analog block, in EE372 you will leverage your knowledge from EE272 and design and fabricate your own digital/analog/mixed-signal chip. This is a completely project-based course where, working in teams, you will propose your own mixed-signal chip, write a Verilog or a synthesizable C++ model of your chip, create a testing/debug strategy for your chip, wrap custom layout to fit into a standard cell system, use synthesis and place and route tools to create the layout of your chip, perform physical verification of your chip and finally tape it out. Useful for anyone who will build a chip in their Ph.D. Pre-requisites: EE271, EE272 and experience in digital/analog circuit design.

EE374 - BLOCKCHAIN FOUNDATIONS

Credits: 3

A detailed exploration of the foundations of blockchains, What blockchains are, how they work, and why they are secure. Transactions, blocks, chains, proof-of-work and stake, wallets, the UTXO model, accounts model, light clients. Throughout the course, students build their own nodes from scratch. Security is defined and rigorously proved. The course is heavy on both engineering and theory. This course is a deeper investigation into the consensus layer of blockchains while CS 251 is a broader investigation, and it can be taken with or without having taken CS 251. Prerequisites: CS106 or equivalent, significant programming experience; CS103 or equivalent; CS109 or EE178 or equivalent.

EE375 - MATH OF MACHINE LEARNING

Credits: 3

Mathematical tools to understand modern machine learning systems. Generalization in machine learning, the classical view: uniform convergence, Radamacher complexity. Generalization from stability. Implicit (algorithmic) regularization. Infinite-dimensional models: reproducing kernel Hilbert spaces. Random features approximations to kernel methods. Connections to neural networks, and neural tangent kernel. Nonparametric regression. Asymptotic behavior of wide neural networks. Properties of convolutional networks. Prerequisites: EE364A or equivalent; Stat310A or equivalent.

EE376 C - UNIVERSAL SCHEMES INFO THEORY

Credits: 3

Universal schemes for lossless and lossy compression, channel coding and decoding, prediction, denoising, and filtering. Characterization of performance limitations in the stochastic setting: entropy rate, rate-distortion function, channel capacity, Bayes envelope for prediction, denoising, and filtering. Lempel-Ziv lossless compression, and Lempel-Ziv based schemes for lossy compression, channel coding, prediction, and filtering. Discrete universal denoising. Compression-based approach to denoising. The compound decision problem. Prerequisites: EE276 (Formerly EE376A).

EE377 - INFORMATION THEORY AND STATS

Credits: 3

Information theoretic techniques in probability and statistics. Fano, Assouad, and Le Cam methods for optimality guarantees in estimation. Large deviations and concentration inequalities (Sanov's theorem, hypothesis testing, the entropy method, concentration of measure). Approximation of (Bayes) optimal procedures, surrogate risks, f-divergences. Penalized estimators and minimum description length. Online game playing, gambling, no-regret learning. Prerequisites: EE 276 (or equivalent) or STATS 300A.

EE378B - INFERENCE, ESTIMATION, INFORM

Credits: 3

Techniques and models for signal, data and information processing, with emphasis on incomplete data, non-ordered index sets and robust low-complexity methods. Linear models; regularization and shrinkage; dimensionality reduction; streaming algorithms; sketching; clustering, search in high dimension; low-rank models; principal component analysis. Applications include: positioning from pairwise distances; distributed sensing; measurement/traffic monitoring in networks; finding communities/clusters in networks; recommendation systems; inverse problems. Prerequisites: EE278 and EE263 or equivalent. Recommended but not required: EE378A

EE376 B - INFORMATION THEORY

Credits: 3

Information theory establishes the fundamental limits on compression and communication over networks. The tools of information theory have also found applications in many other fields, including probability and statistics, computer science and physics. The course will cover selected topics from these applications, including communication networks, through regular lectures and student projects. Prerequisites: EE276 (Formerly EE376A)

EE376 D - WIRELESS INFORMATION THEORY

Credits: 3

Information theory forms the basis for the design of all modern day communication systems. The original theory was primarily point-to-point, studying how fast information can flow across an isolated noisy communication channel. Until recently, there has been only limited success in extending the theory to a network of interacting nodes. Progress has been made in the past decade driven by engineering interest in wireless networks. The course provides a unified overview of this recent progress made in information theory of wireless networks. Starting with an overview of the capacity of fading and multiple-antenna wireless channels, we aim to answer questions such as: What is the optimal way for users to cooperate and exchange information in a wireless network? How much benefit can optimal cooperation provide over traditional communication architectures? How can cooperation help to deal with interference between multiple wireless transmissions? Prerequisites: EE276 (Formerly EE376A).

EE378A - STATISTICAL SIGNAL

Credits: 3

Basic concepts of statistical decision theory; Bayes decision theory; HMMs and their state estimation (Forward-backward), Kalman as special case, approximate state estimation (particle filtering, Extended Kalman Filter), unknown parameters; Inference under logarithmic loss, mutual information as a fundamental measure of statistical relevance, properties of mutual information: data processing, chain rules. Directed information. Prediction under logarithmic loss; Context Tree Weighting algorithm; Sequential decision making in general: prediction under general loss functions, causal estimation, estimation of directed information. Non-sequential inference via sequential probability assignments. Universal denoising; Denoising from a decision theoretic perspective: nonparametric function estimation, wavelet shrinkage, density estimation; Estimation of mutual information on large alphabets with applications such as boosting the Chow-Liu algorithm. Estimation of the total variation distance, estimate the fundamental limit is easier than to achieve the fundamental limit; Peetre's K-functional and bias analysis: bias correction using jackknife, bootstrap, and Taylor series; Nonparametric functional estimation. Prerequisites: Familiarity with probability theory and linear algebra at the undergraduate level.

EE378C - INFORMATION LOWER BOUNDS

Credits: 3

Ideas and techniques for information-theoretic lower bounds, with examples in machine learning, statistics, information theory, theoretical computer science, optimization, online learning and bandits, operations research, and more. Deficiency and Le Cam's distance; classical asymptotics; information measures and joint range; Le Cam, Assouad, and Fano; Ingster-Suslina method; method of moments; strong converses; constrained risk inequality; compression arguments; privacy-constrained estimation; sequential experimental design; statistical/computational tradeoff Prerequisites: EE 278, CS 229T, STATS 300A, or equivalent, or instructor's permission.

EE379 - DIGITAL COMMUNICATION

Credits: 3

Modulation: linear, differential and orthogonal methods; signal spaces; power spectra; bandwidth requirements. Detection: maximum likelihood and maximum a posteriori probability principles; sufficient statistics; correlation and matched-filter receivers; coherent, differentially coherent and noncoherent methods; error probabilities; comparison of modulation and detection methods. Intersymbol interference: single-carrier channel model; Nyquist requirement; whitened matched filter; maximum likelihood sequence detection; Viterbi algorithm; linear equalization; decision-feedback equalization. Multi-carrier modulation: orthogonal frequency-division multiplexing; capacity of parallel Gaussian channels; comparison of single- and multi-carrier techniques. Prerequisite: EE102B and EE278 (or equivalents). EE279 is helpful but not required.

EE382A - PARALLEL PROC BEYOND MULTICORE

Credits: 3

Formerly EE392Q. The current parallel computing research emphasizes multi-cores, but there are alternative array processors with significant potential. This hands-on course focuses on SIMD (Single-Instruction, Multiple-Data) massively parallel processors. Topics: Flynn's Taxonomy, parallel architectures, Kestrel architecture and simulator, principles of SIMD programming, parallel sorting with sorting networks, string comparison with dynamic programming (edit distance, Smith-Waterman), arbitrary-precision operations with fixed-point numbers, reductions, vector and matrix multiplication, image processing algorithms, asynchronous algorithms on SIMD ("SIMD Phase Programming Model"), Man-delbrot set, analysis of parallel performance.

EE384A - INTERNET PROTOCOLS/STANDARDS

Credits: 3

Local area networks addressing and switching; IEEE 802.1 bridging protocols (transparent bridging, virtual LANs). Internet routing protocols: interior gateways (RIP, OSPF) and exterior gateways (BGP); multicast routing; multiprotocol label switching (MPLS). Routing in mobile networks: Mobile IP, Mobile Ad Hoc Networks (MANET), Wireless Mesh Networks. Prerequisite: EE 284 or CS 144.

EE384E - WIRELESS NETWORKING

Credits: 3

Design and implementation of wireless networks and mobile systems. The course will commence with a short retrospective of wireless communication and initially touch on some of the fundamental physical layer properties of various wireless communication technologies. The focus will then shift to design of media access control and routing layers for various wireless systems. The course will also examine adaptations necessary at transport and higher layers to cope with node mobility and error-prone nature of the wireless medium. Finally, it will conclude with a brief overview of other related issues including emerging wireless/mobile applications. Prerequisites: EE 284

EE380 - COLLOQUIUM ON COMPUTER SYSTEMS

Credits: 999

Live presentations of current research in the design, implementation, analysis, and applications of computer systems. Topics range over a wide range and are different every quarter. Topics may include fundamental science, mathematics, cryptography, device physics, integrated circuits, computer architecture, programming, programming languages, optimization, applications, simulation, graphics, social implications, venture capital, patent and copyright law, networks, computer security, and other topics of related to computer systems. May be repeated for credit.

EE382C - INTERCONNECTION NETWORKS

Credits: 3

The architecture and design of interconnection networks used to communicate from processor to memory, from processor to processor, and in switches and routers. Topics: network topology, routing methods, flow control, router microarchitecture, and performance analysis. Enrollment limited to 30. Prerequisite: 282.

EE384C - WIRELESS LANS

Credits: 3

Characteristics of wireless communication: multipath, noise, and interference. Communications techniques: spread-spectrum, CDMA, and OFDM. IEEE 802.11 physical layer specifications: FHSS, DSSS, IEEE 802.11b (CCK), and 802.11a/g (OFDM). IEEE 802.11 media access control protocols: carrier sense multiple access with collision avoidance (CSMA/CA), point coordination function (PCF), IEEE 802.11e for differentiated services. IEEE 802.11 network architecture: ad hoc and infrastructure modes, access point functionality. Management functions: synchronization, power management and association. IEEE 802.11s Mesh Networks. IEEE 802.16 (WiMAX) network architecture and protocols: Physical Layer (OFDMA) and Media Access Control Layer. Current research papers in the open literature. Prerequisite: EE 284 or CS 244A.

EE384S - PERFORMANCE ENGINEERING OF COM

Credits: 3

Modeling and control methodologies for high-performance network engineering, including: Markov chains and stochastic modeling, queueing networks and congestion management, dynamic programming and task/processor scheduling, network dimensioning and optimization, and simulation methods. Applications for design of high-performance architectures for wireline/wireless networks and the Internet, including: traffic modeling, admission and congestion control, quality of service support, power control in wireless networks, packet scheduling in switches, video streaming over wireless links, and virus/worm propagation dynamics and countermeasures. Enrollment limited to 30. Prerequisites: basic networking technologies and probability.

EE385A - ROBUST AND TESTABLE SYSTEMS

Credits: 999

Student/faculty discussions of research problems in the design of reliable digital systems. Areas: fault-tolerant systems, design for testability, production testing, and system reliability. Emphasis is on student presentations and Ph.D. thesis research. May be repeated for credit. Prerequisite: consent of instructor.

EE388 - MODERN CODING THEORY

Credits: 3

Tools for analysis and optimization of iterative coding systems. LDPC, turbo and, RA codes. Optimized ensembles, message passing algorithms, density evolution, and analytic techniques. Prerequisite: EE 276.

EE391 - SPECIAL STUDIES/REPORTS IN EE

Credits: 999

Independent work under the direction of a faculty member; written report or written examination required. Letter grade given on the basis of the report; if not appropriate, student should enroll in 390. May be repeated for credit.

EE392 B - INDUSTRIAL AI

Credits: 999

The seminar features guest lectures from the industry. The Industrial AI (I-AI) computing applications are at the center of on-going digital transformation. Known as the Fourth Industrial Revolution, or Industry 4.0, this is a multi-trillion-dollar transformation of economy. The I-AI is related to Internet of Things (IoT), where 'things' include man-made systems and business processes: industrial, transportation, operations and support, and supply chains. I-AI applications are mission critical with large cost of error compared to AI apps for the Internet of People. The lecturers from technology (e.g., computing) companies, consultancies, AI vendors, OEMs, and end users of the I-AI will discuss business and 'big picture' technical issues. Example vertical industries are energy, transportation, oil and gas, data centers, and manufacturing.

EE387 - ALGEBRAIC ERROR CORRECTING COD

Credits: 3

Introduction to the theory of error correcting codes, emphasizing algebraic constructions, and diverse applications throughout computer science and engineering. Topics include basic bounds on error correcting codes; Reed-Solomon and Reed-Muller codes; list-decoding, list-recovery and locality. Applications may include communication, storage, complexity theory, pseudorandomness, cryptography, streaming algorithms, group testing, and compressed sensing. Prerequisites: Linear algebra, basic probability (at the level of, say, CS109, CME106 or EE178) and "mathematical maturity" (students will be asked to write proofs). Familiarity with finite fields will be helpful but not required.

EE390 - SPECIAL STUDIES OR PROJS IN EE

Credits: 999

Independent work under the direction of a faculty member. Individual or team activities may involve lab experimentation, design of devices or systems, or directed reading. May be repeated for credit.

EE392 AA - DATA TRANSMISSION

Credits: 3

EE 3392AA focuses on state-of-the-art data communication system theory and design, particularly systems with multiple users and dimensions (MIMO over parallel antennas or wires). The focus is on multi-user physical-layer channels like multiple access, broadcast, and interference channels, their capacity regions and designs to achieve any points therein. Examples include the latest cellular, Wi-Fi, wireline, cable, and other systems that stress fundamental transmission limits. Topics include system design, particularly physical-layer modulation/coding analysis and optimization through various artificial intelligence optimization methods for multi-dimensional channels. Included are methods to design and adapt both transmitter and receiver to variable channels. Prerequisites: EE 278, linear algebra, EE 279 or EE 379, or instructor consent.

EE392 F - LARGE SCALE CONVEX OPTIMI

Credits: 3

This course presents a unified analysis of large-scale convex optimization algorithms through the abstraction of monotone operators. The topics include monotone operators, primal-dual methods, randomized coordinate update methods, ADMM-type methods, maximality, duality, acceleration, scaled relative graphs, and distributed and decentralized optimization

EE392I - TRENDS IN COMPUTING

Credits: 1

Lectures series and invited talks on current trends in computing and communications, and ongoing initiatives for research and open innovation. This year's focus on evolving cloud computing architectures and their impact on the enterprise; big data trends and rise of the third platform; software as a service; wireless and cellular network architectures; mobility and mobile data proliferation; open mobile platforms (e.g. Android); multi-homed mobile networking, associated data communication and mobile resource trade-offs, and system implementation in smartphones and Android devices.

EE392T - SEMINAR IN CHIP TEST AND DEBUG

Credits: 1

Seminars by industry professionals in digital IC manufacturing test and silicon debug. Topics include yield and binsplit modeling, defect types and detection, debug hardware, physical analysis, and design for test/debug circuits. Case studies of silicon failures. Prerequisite: basic digital IC design (271 or 371).

EE402A - TOPICS INT'L TECH MANAGEMENT

Credits: 999

Autumn 2022 Theme: "New Business in Asia Driven by Climate-Tech and Sustainability." Guest speakers from industry and academia present examples of problem-driven innovation for environmental sustainability, climate change amelioration and mitigation, ESG reporting, SDG targeting, and related topics. Focus is on present-day developments in Asia business, technology, and investing, along with their implications for Silicon Valley and the U.S.

EE42 - INTRO ENGINEERING ELECTROMAGNE

Credits: 5

Electricity and magnetism and its essential role in modern electrical engineering devices and systems, such as sensors, displays, DVD players, and optical communication systems. The topics that will be covered include electrostatics, magnetostatics, Maxwell's equations, one-dimensional wave equation, electromagnetic waves, transmission lines, and one-dimensional resonators. Pre-requisites: none.

EE469B - RF PULSE DESIGN FOR MRI

Credits: 3

Magnetic resonance imaging (MRI) and spectroscopy (MRS) based on the use of radio frequency pulses to manipulate magnetization. Analysis and design of major types of RF pulses in one and multiple dimensions, analysis and design of sequences of RF pulses for fast imaging, and use of RF pulses for the creation of image contrast in MRI. Prerequisite: 369B.

EE392K - SELF-PROGRAMMING NETWORKS

Credits: 3

This is an advanced topics course on building autonomous networks using data and techniques from machine learning. It covers two major application areas: Cloud Computing Systems and Mobile Wireless Networks. The course introduces the architecture of Self-Programming Networks for sensing, inferring, learning and control, consisting of (i) a "reflex layer" for inferring at line rate and at scale, and (ii) a "deliberate layer" for efficient resource scheduling and network control. Various sensing and inference algorithms for deriving insights and alerts from the sensed data will be discussed. Methods for synchronizing clocks across a large data center and using this to reconstruct the fine details of network performance (queue-depths, link utilizations and buffer and link compositions) in near real-time will be presented. Similarly, methods for inferring available bandwidth in dynamic mobile networks and using it to drive different application optimizations will be presented. Students will learn the use of neural networks and learning techniques (a) to accelerate inference and control algorithms, (b) for "workload fingerprinting", (c) for predicting wireless link capacities, and (d) for scheduling resources. Finally, the principles of creating an interactive database for detecting anomalies, raising alerts, and serving insights to the user will be discussed. The course involves a team-based project.

EE400 - THESIS RESEARCH

Credits: 999

Limited to candidates for the degree of Engineer or Ph.D. May be repeated for credit.

EE402T - ENTREPRENEURSHIP IN ASIA

Credits: 999

Distinctive patterns and challenges of entrepreneurship in Asia; update of business and technology issues in the creation and growth of start-up companies in major Asian economies. Distinguished speakers from industry, government, and academia.

EE46 - ENGINEERING FOR GOOD

Credits: 3

Projects that provide immediate and positive impact on the world. Focus is on global health and sustainable development by learning from experts in these fields. Students work on real-world projects with help from members of NGOs and social entrepreneurial companies as part of the hands-on learning experience. Prerequisite: ENGR 21 or ENGR 40M or EE 122A or CS 106B or consent of instructor.

EE60N - MAN VERSUS NATURE

Credits: 4

Preference to freshman. Natural hazards, earthquakes, volcanoes, floods, hurricanes, and fires, and how they affect people and society; great disasters such as asteroid impacts that periodically obliterate many species of life. Scientific issues, political and social consequences, costs of disaster mitigation, and how scientific knowledge affects policy. How spaceborne imaging technology makes it possible to respond quickly and mitigate consequences; how it is applied to natural disasters; and remote sensing data manipulation and analysis. GER:DB-EngrAppSci

EE64SI - MECHANICAL PROTOTYPING

Credits: 2

This course will give non-mechanical engineers experience designing mechanical assemblies specifically for manufacture by readily accessible tools, such as 3-D printers and laser cutters. It will also teach students to debug their own mechanical designs, and interface them with other components (such as store-bought parts). By the end of the quarter students will feel comfortable independently designing and manufacturing simple assemblies to serve useful functions in their lives.

EE801 - TGR PROJECT

Credits: 999

May be repeated for credit.

EE84N - INTERNET OF THINGS

Credits: 3

Driven by the ubiquity of the Internet and advances in various technological fields, all aspects of the physical world in which we live are undergoing a major transformation. Underlying this transformation is a concept known as the Internet of Things (IoT) which envisions that every physical object in the world could be connected to the Internet. This concept is at the root of such developments as the fourth industrial revolution, precision agriculture, smart cities, intelligent transportation, home and building automation, precision medicine, etc. In this seminar, we trace back the origins of the IoT concept in terms of both the vision and pioneering work, identify the building blocks of an IoT system, and explore enabling technologies pertaining to the devices that get attached to things (possibly comprising sensors, actuators, and embedded systems) and the communications capabilities (RFID, Bluetooth, wireless sensor networks, Wi-Fi, Low Power WANs, cellular networks, vehicular communications). Students will apply the acquired knowledge to the design of IoT systems meeting specific objectives in various application domains.

EE65 - MODERN PHYSICS FOR ENGINEERS

Credits: 4

This course introduces the core ideas of modern physics that enable applications ranging from solar energy and efficient lighting to the modern electronic and optical devices and nanotechnologies that sense, process, store, communicate and display all our information. Though the ideas have broad impact, the course is widely accessible to engineering and science students with only basic linear algebra and calculus through simple ordinary differential equations as mathematics background. Topics include the quantum mechanics of electrons and photons (Schrödinger's equation, atoms, electrons, energy levels and energy bands; absorption and emission of photons; quantum confinement in nanostructures), the statistical mechanics of particles (entropy, the Boltzmann factor, thermal distributions), the thermodynamics of light (thermal radiation, limits to light concentration, spontaneous and stimulated emission), and the physics of information (Maxwell's demon, reversibility, entropy and noise in physics and information theory). Pre-requisite: Physics 41. Pre- or co-requisite: Math 53 or CME 102.

EE802 - TGR DISSERTATION

Credits: 999

May be repeated for credit.